

EE 116

FPGA-Based Digital System Design

Lab Tutorial

School of Information Science and Technology

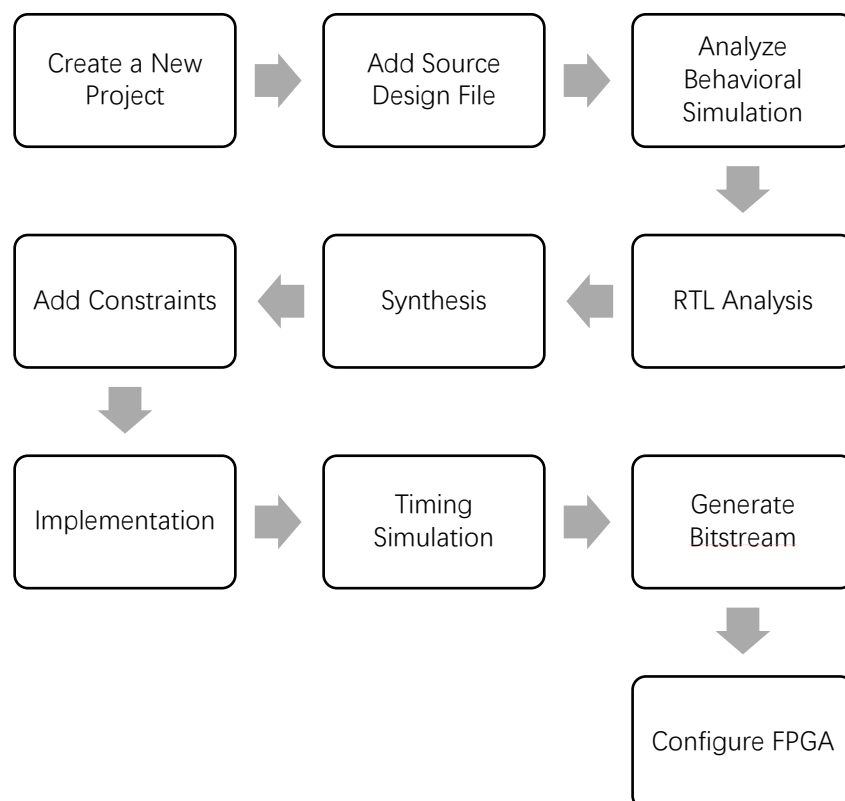
ShanghaiTech University

Sept. 26, 2021

## Tutorial 2 Vivado + Zedboard Development Tutorial

### Introduction

In our lab we will guide you to create a simple digital design using Xilinx Vivado software. A typical design flow consists of creating new projects, adding source design file, analyzing behavioral simulation, RTL analysis, synthesis, adding constraints, implementation, timing simulation, generating bitstream and configuring FPGA. You will go through the typical design flow targeting the Zynq-7000 All Programmable SoC based ZedBoard. The typical design flow is shown below. We focus on the last seven steps in this tutorial.



**A typical design flow**

### Objective:

Achieve the traditional FPGA development under the environment of Vivado. Here we only consider the PL development. In this experiment, we use switches to control LEDs.

**Environment:** Vivado 2020.2

**Development board:** Zedboard xc7z020clg484-1

## Main Flow:

**Step1: Create a New project.**

**Step2: Add Source.**

**Step3: Behavioral Simulation**

**Step4: RTL Analysis**

**Step5: Synthesis**

**Step6: Add constraints**

**Step7: Implementation**

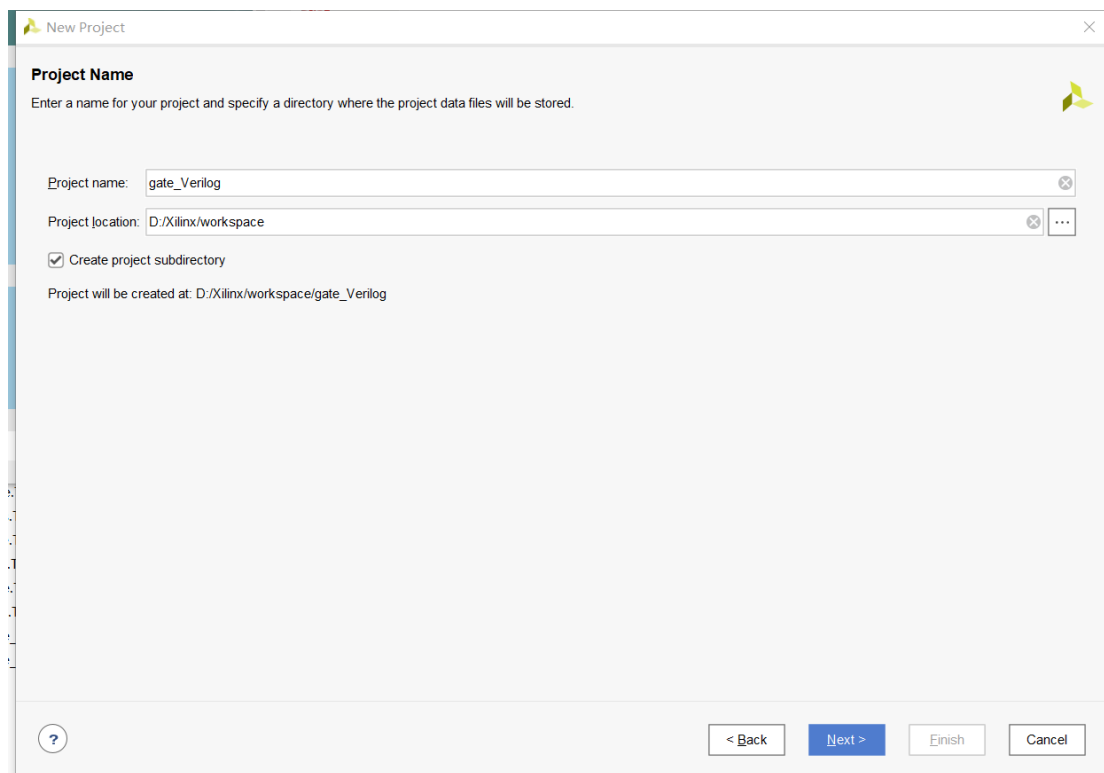
**Step8: Run Post-Implementation Timing Simulation**

**Step9: Generate Bitstream**

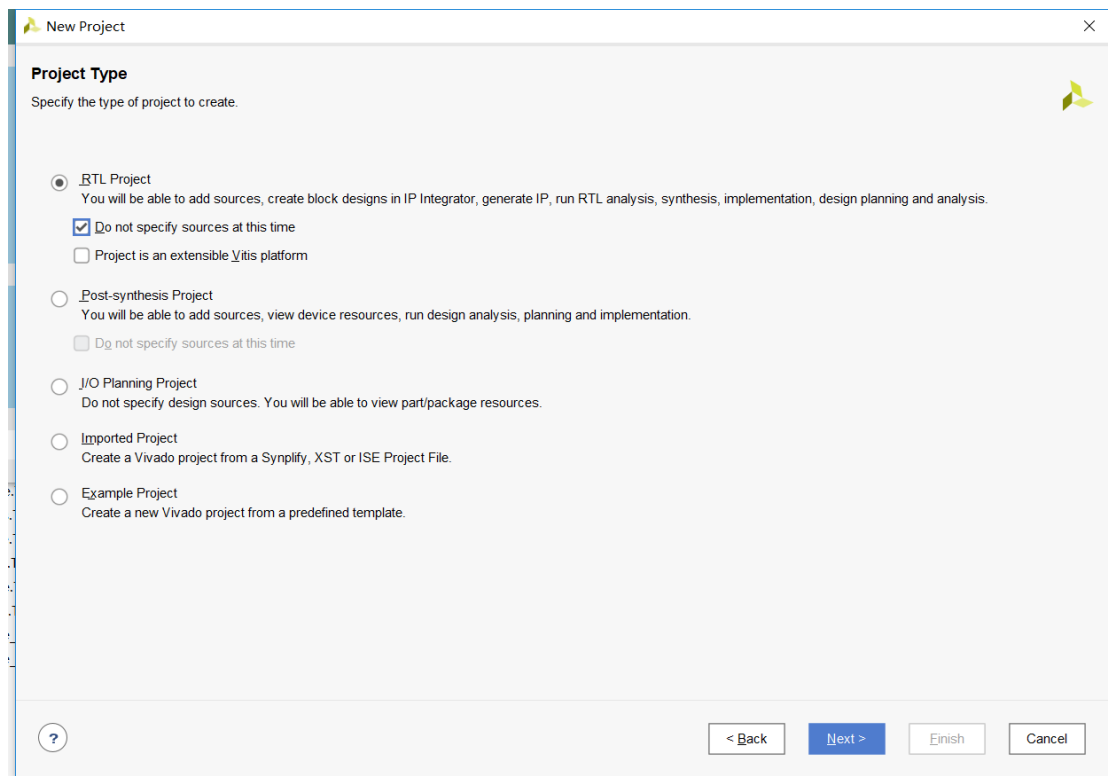
**Step10: Connect Zedboard and Configure FPGA,**

### Step1: Create a New project.

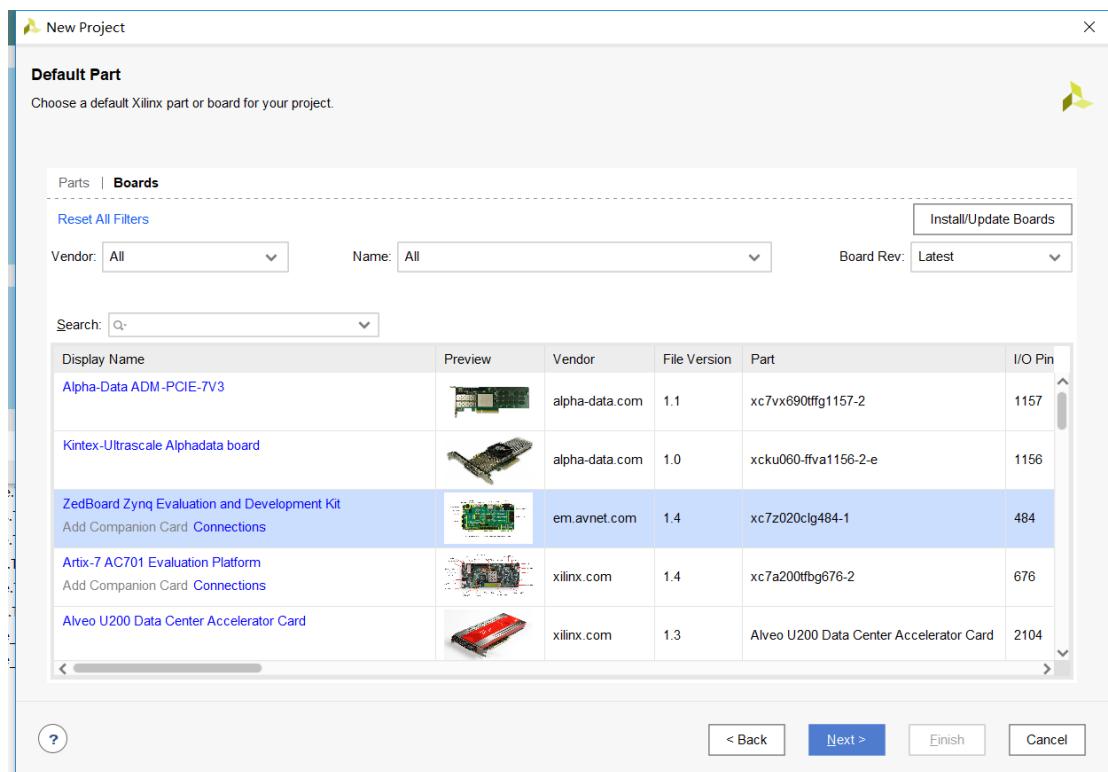
Open Vivado 2020.2. and build a new project., click “**Create a New Vivado Project**”, then pop out dialog box “**Project Name**”, write your own project name, here we use “**gate\_Verilog**”, then click “**Next**”.

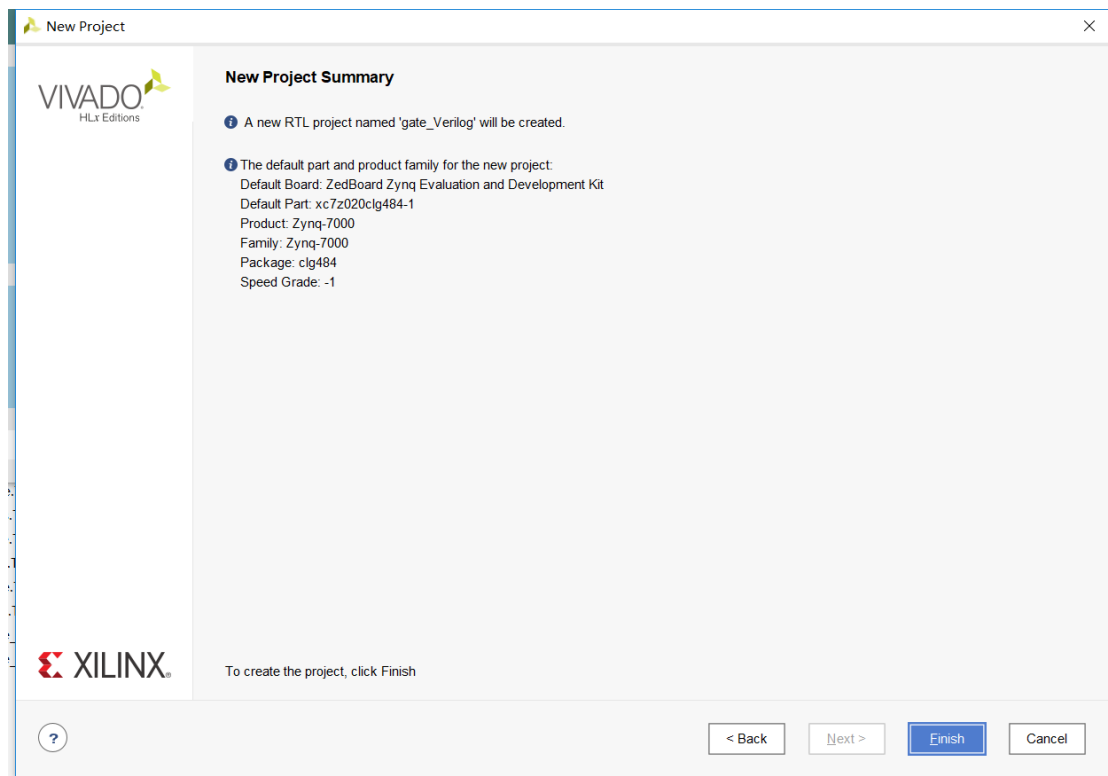


Then we select “**RTL Project**”, in the meantime we select “**Do not specify sources at this time**”, then click **Next**.



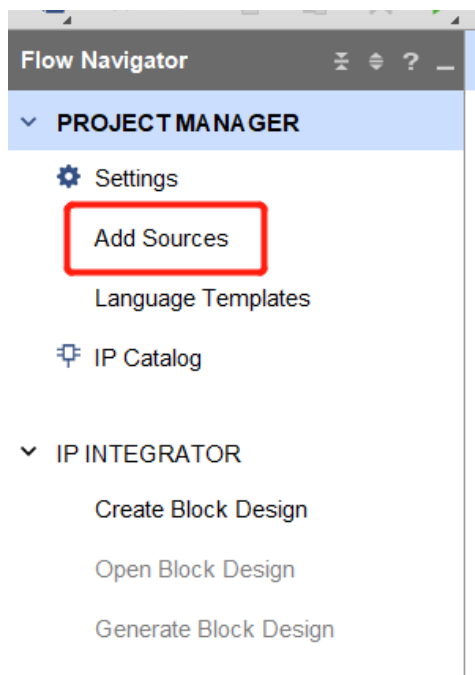
Here click **“Boards”**. and select **“ZedBoard Zynq Evaluation and Development Kit”** board. then click **“Next”**, and click **“Finish”**.



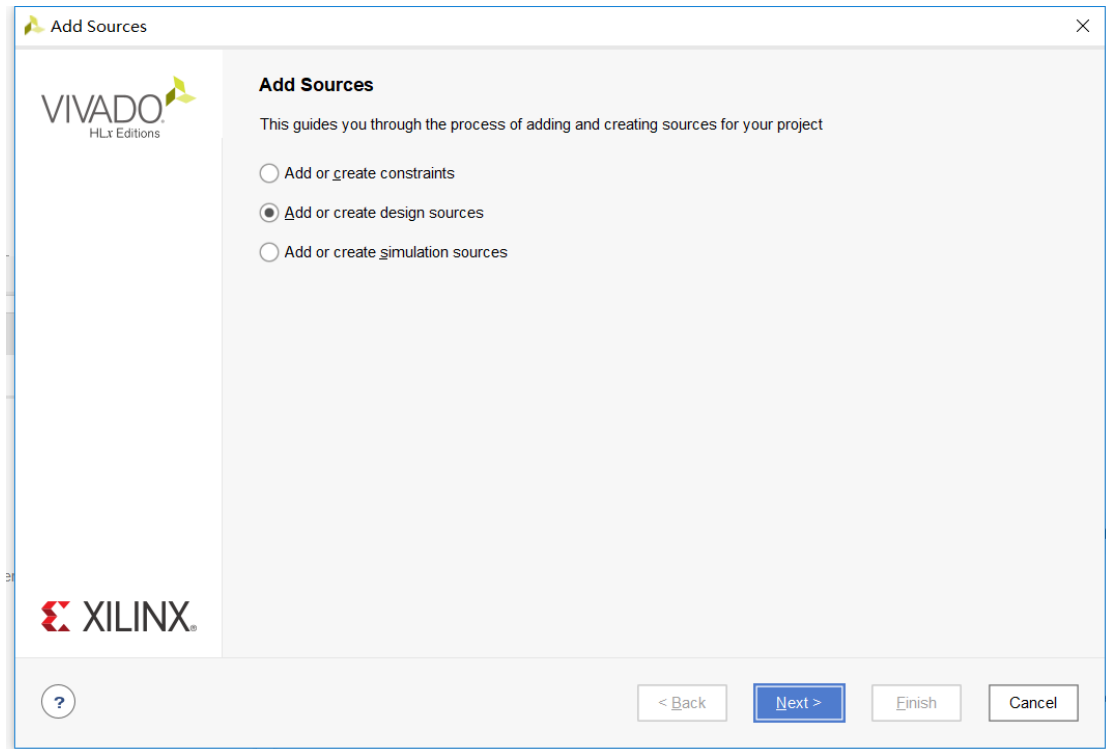


## Step2: Add Source.

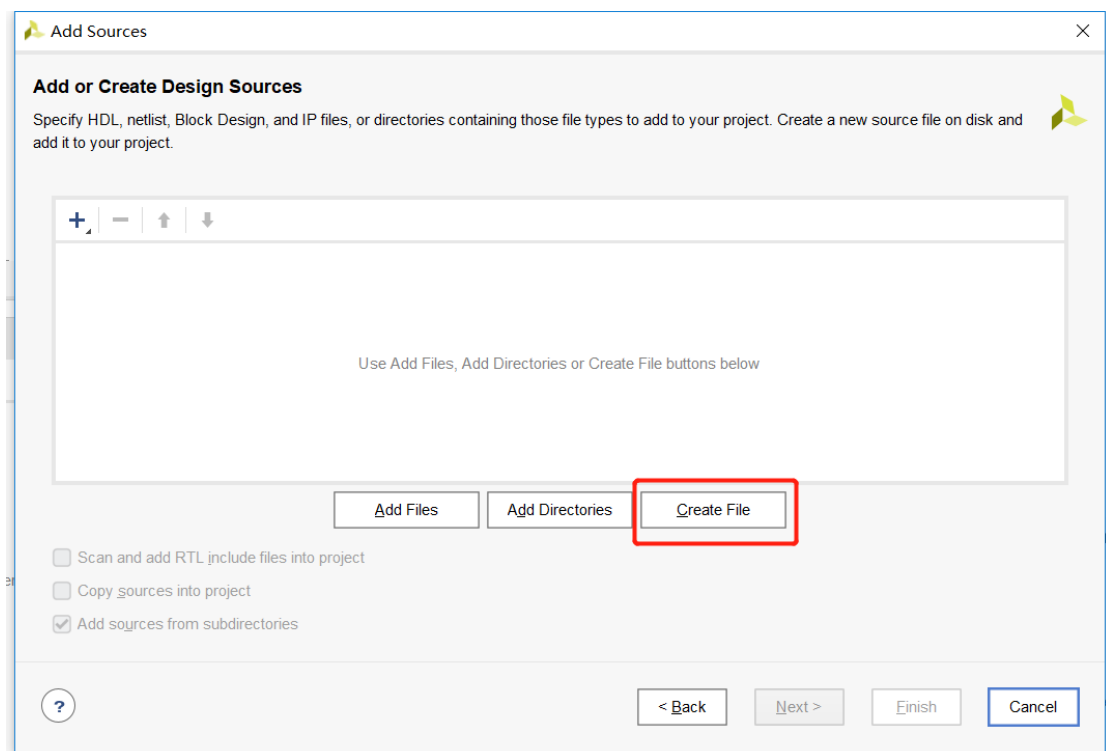
From left window “**Flow Navigator**”, Click “**Add Source**”,



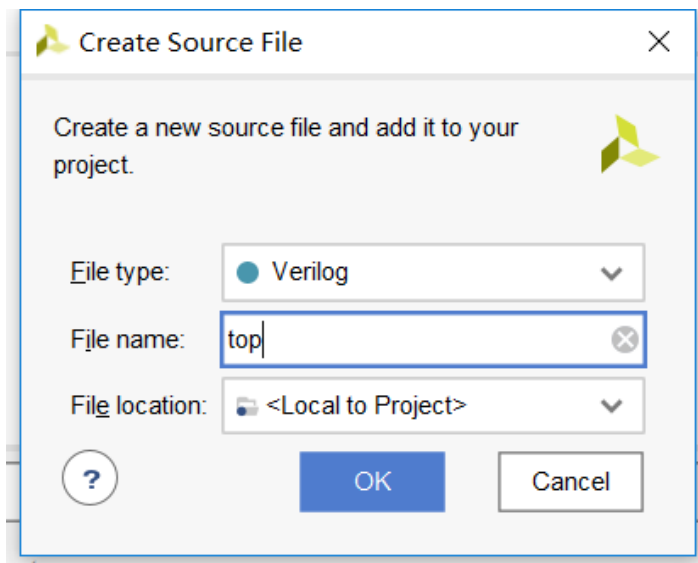
Select “**Add or create design sources**”, then Click “**Next**”,



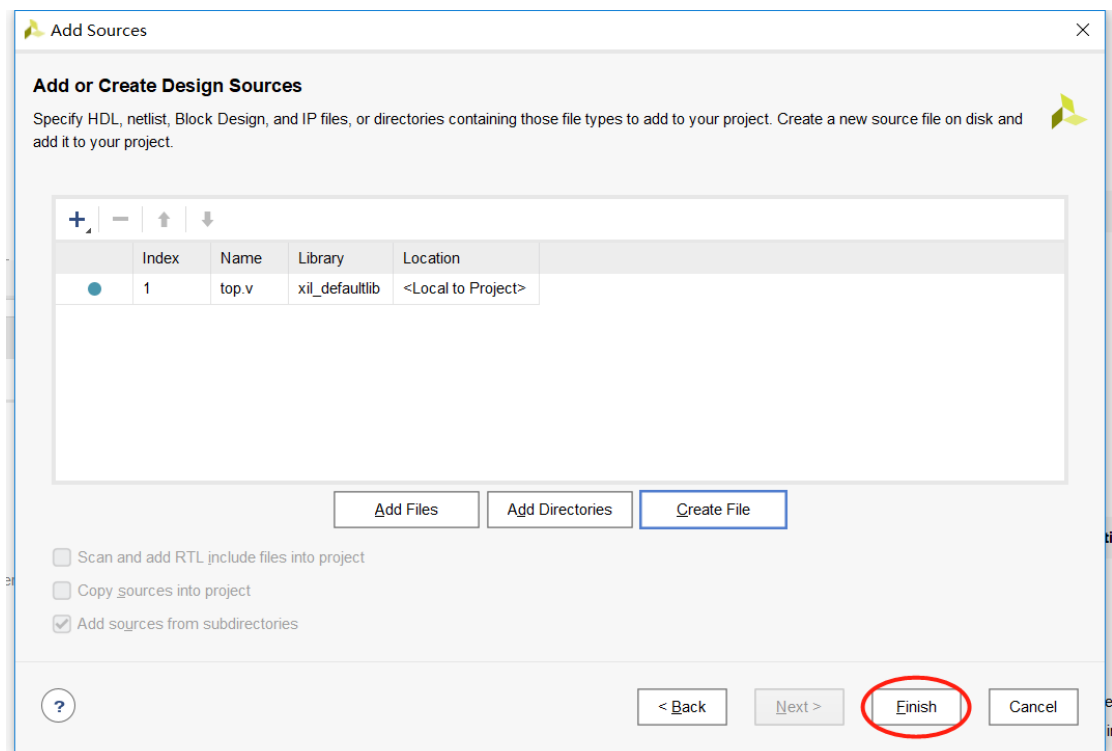
Click “Create File”,



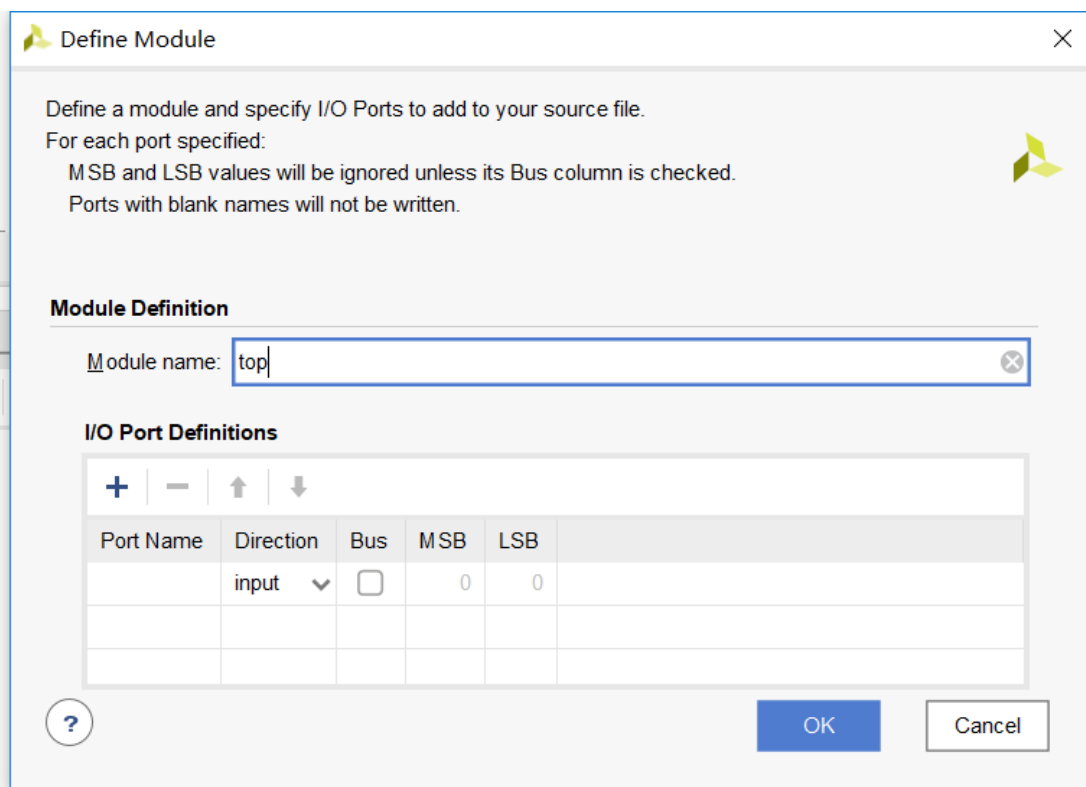
Import the “File name”, “File type” selects “Verilog”. Click “OK”,



Click **"Finish"**



Pop out the dialog box **"Define Module"**, Click **"OK"**,



Define a module and specify I/O Ports to add to your source file.  
For each port specified:  
MSB and LSB values will be ignored unless its Bus column is checked.  
Ports with blank names will not be written.

**Module Definition**

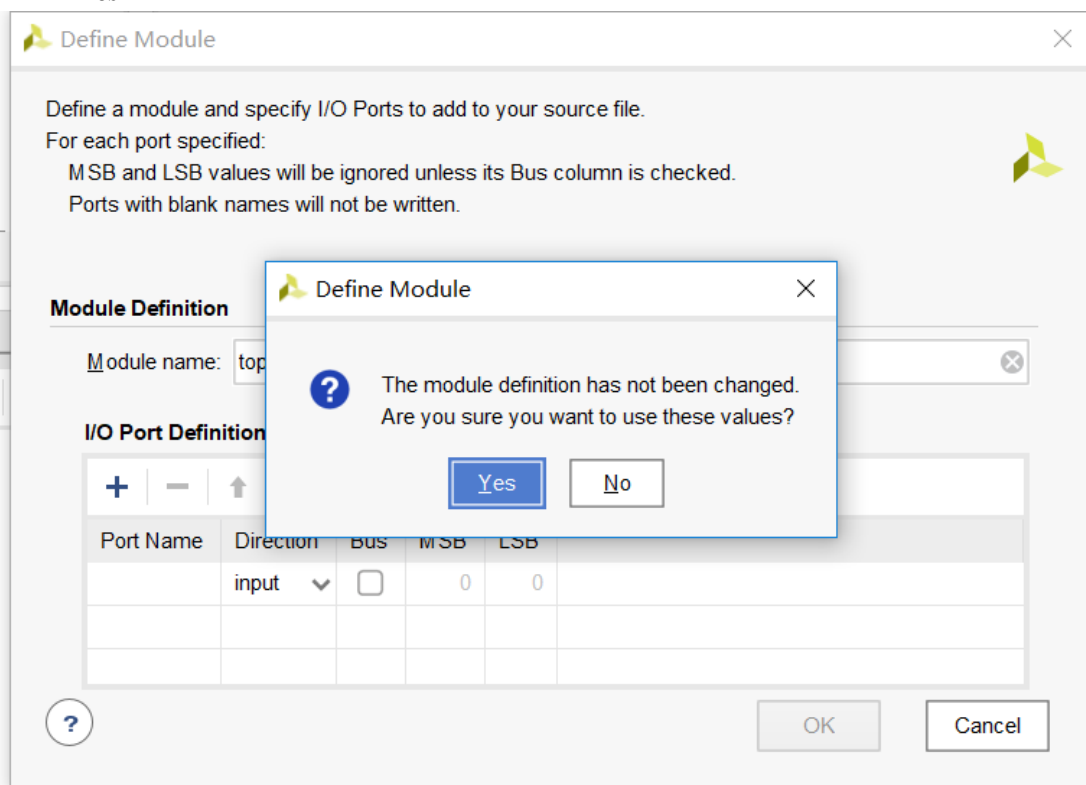
Module name:

**I/O Port Definitions**

| Port Name | Direction | Bus                      | MSB | LSB |
|-----------|-----------|--------------------------|-----|-----|
|           | input     | <input type="checkbox"/> | 0   | 0   |
|           |           |                          |     |     |
|           |           |                          |     |     |

OK Cancel

Click “Yes”



Define a module and specify I/O Ports to add to your source file.  
For each port specified:  
MSB and LSB values will be ignored unless its Bus column is checked.  
Ports with blank names will not be written.

**Module Definition**

Module name:

**I/O Port Definition**

| Port Name | Direction | Bus                      | MSB | LSB |
|-----------|-----------|--------------------------|-----|-----|
|           | input     | <input type="checkbox"/> | 0   | 0   |
|           |           |                          |     |     |
|           |           |                          |     |     |

OK Cancel

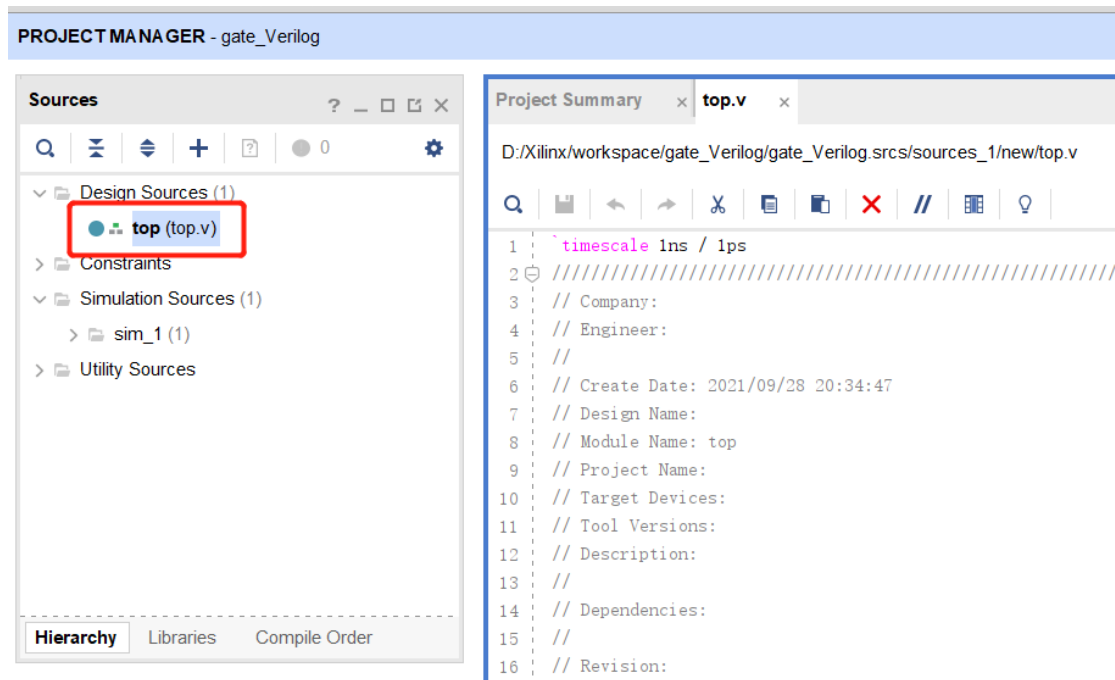
**Define Module**

The module definition has not been changed.  
Are you sure you want to use these values?

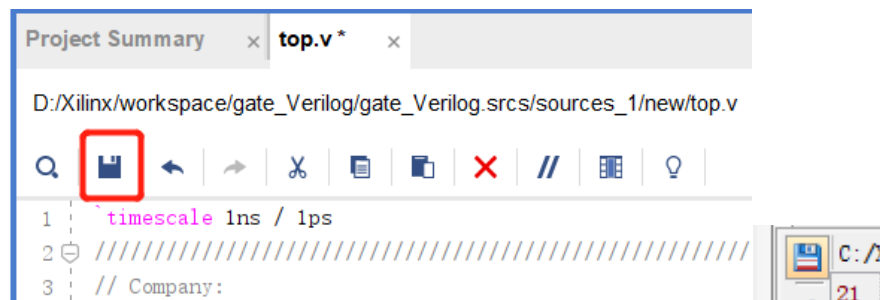
Yes No

Find Source Window. Double click “**top (top.v)**”. Open the edit box.





Write Verilog code, Click “save”



**top.v**

```

module top(
    input a,
    input b,
    output [5:0] z
);
    assign z[0] = a & b;
    assign z[1] = !(a & b);
    assign z[2] = a | b;
    assign z[3] = !(a | b);
    assign z[4] = a ^ b;
    assign z[5] = !(a ^ b);

endmodule

```

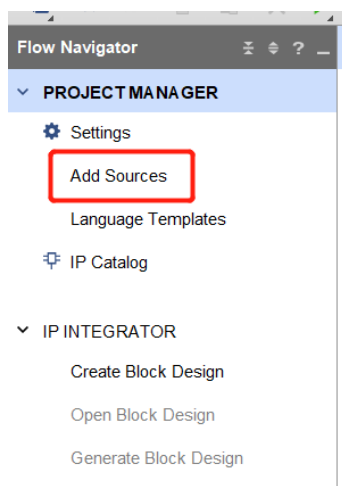
```

5 //
6 // Create Date: 2021/09/28 20:34:47
7 // Design Name:
8 // Module Name: top
9 // Project Name:
10 // Target Devices:
11 // Tool Versions:
12 // Description:
13 //
14 // Dependencies:
15 //
16 // Revision:
17 // Revision 0.01 - File Created
18 // Additional Comments:
19 //
20 ///////////////////////////////////////////////////////////////////
21
22
23 module top(
24     input a,
25     input b,
26     output [5:0] z
27 );
28
29     assign z[0] = a & b;
30     assign z[1] = !(a & b);
31     assign z[2] = a | b;
32     assign z[3] = !(a | b);
33     assign z[4] = a ^ b;
34     assign z[5] = !(a ^ b);
35
36 endmodule
37

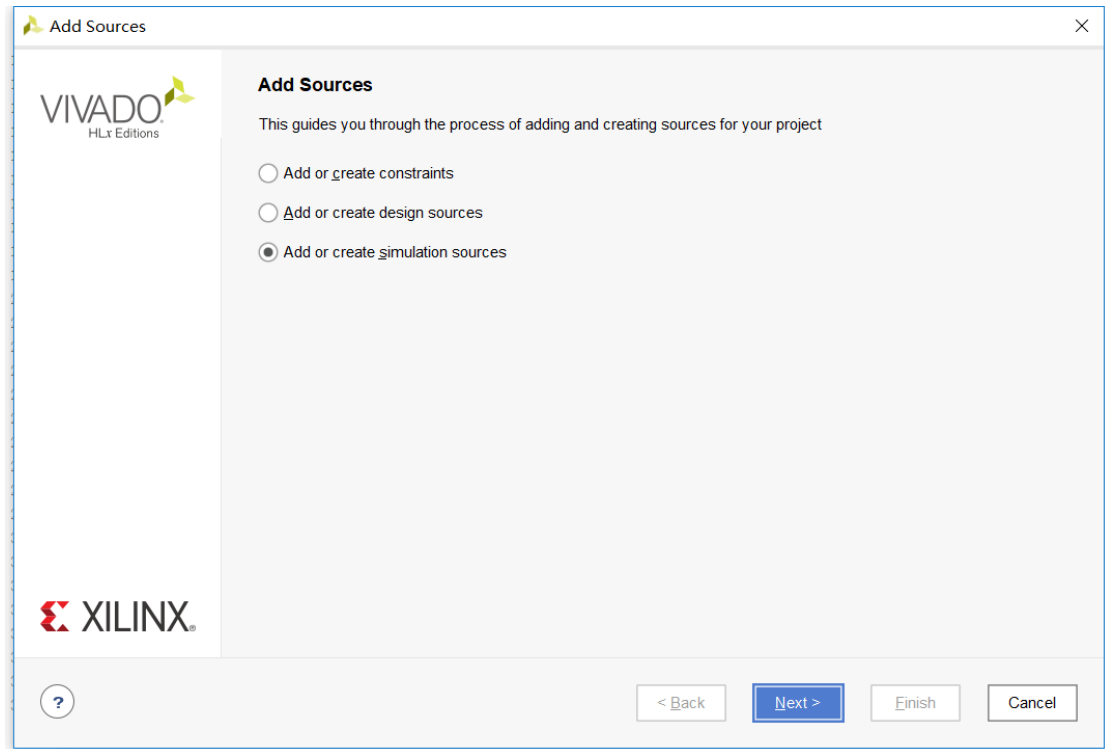
```

### Step3: Behavioral Simulation

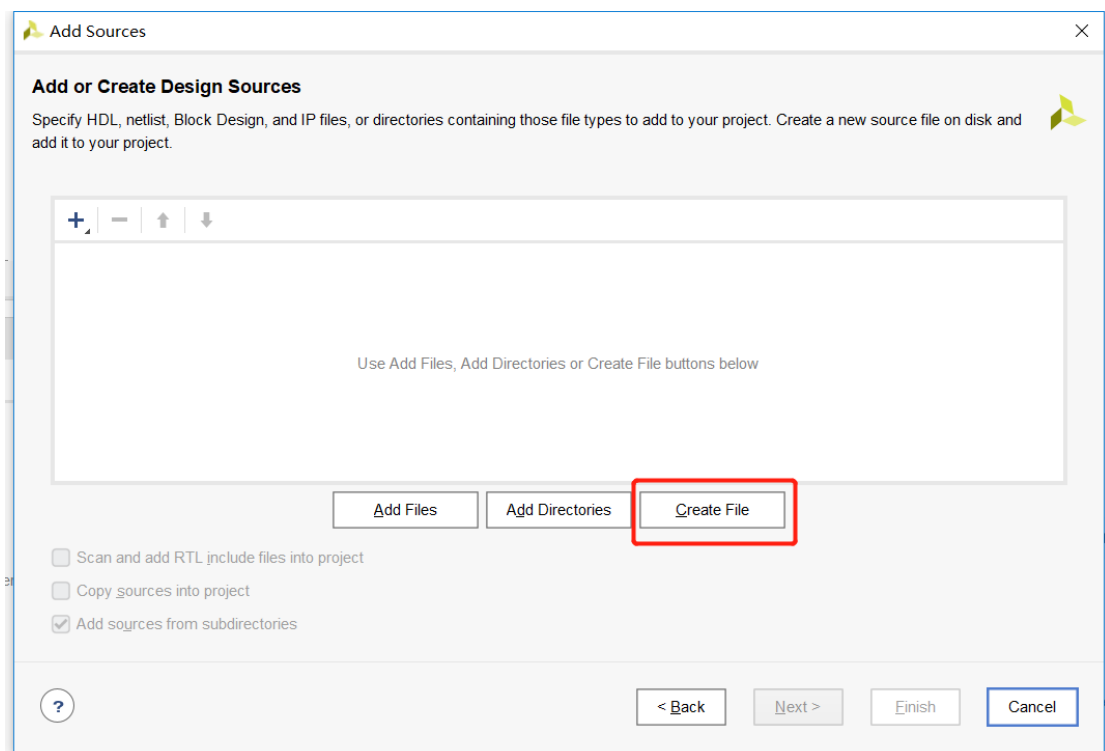
Click “Add Sources”



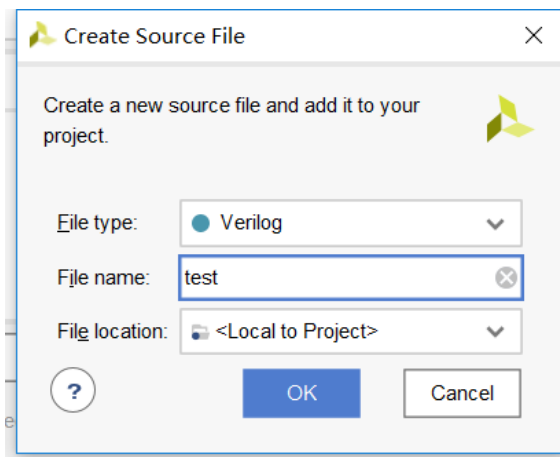
Select “Add or create simulation sources”, Click “Next”



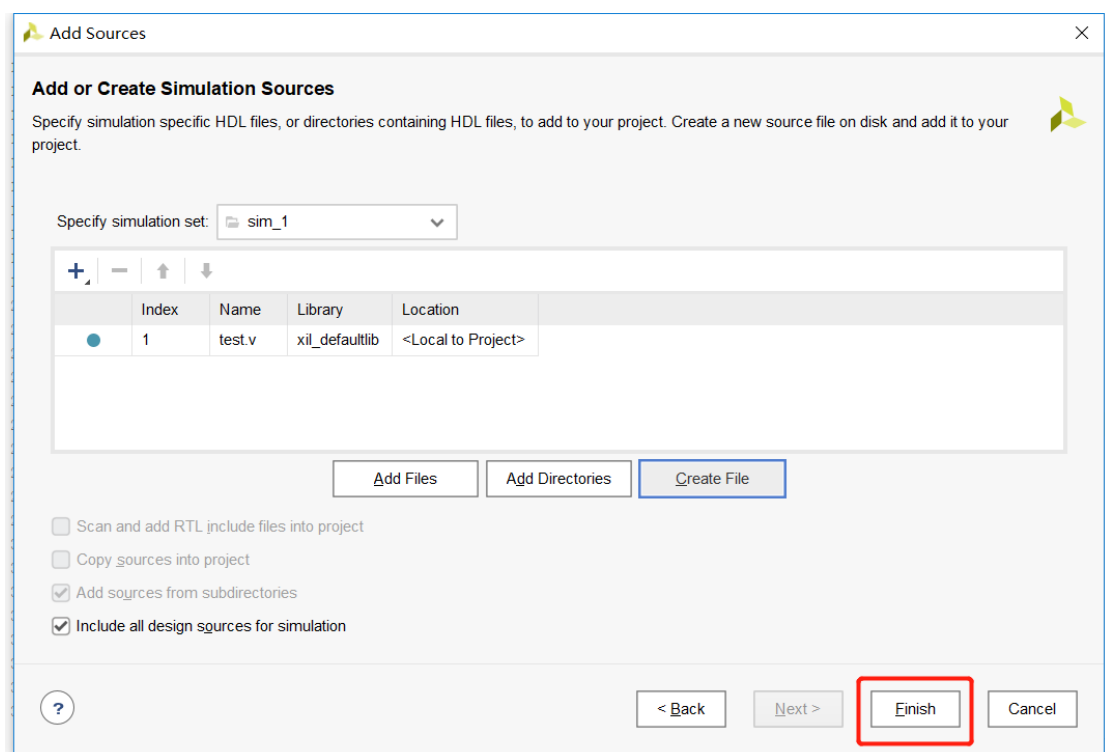
Click “Create File”



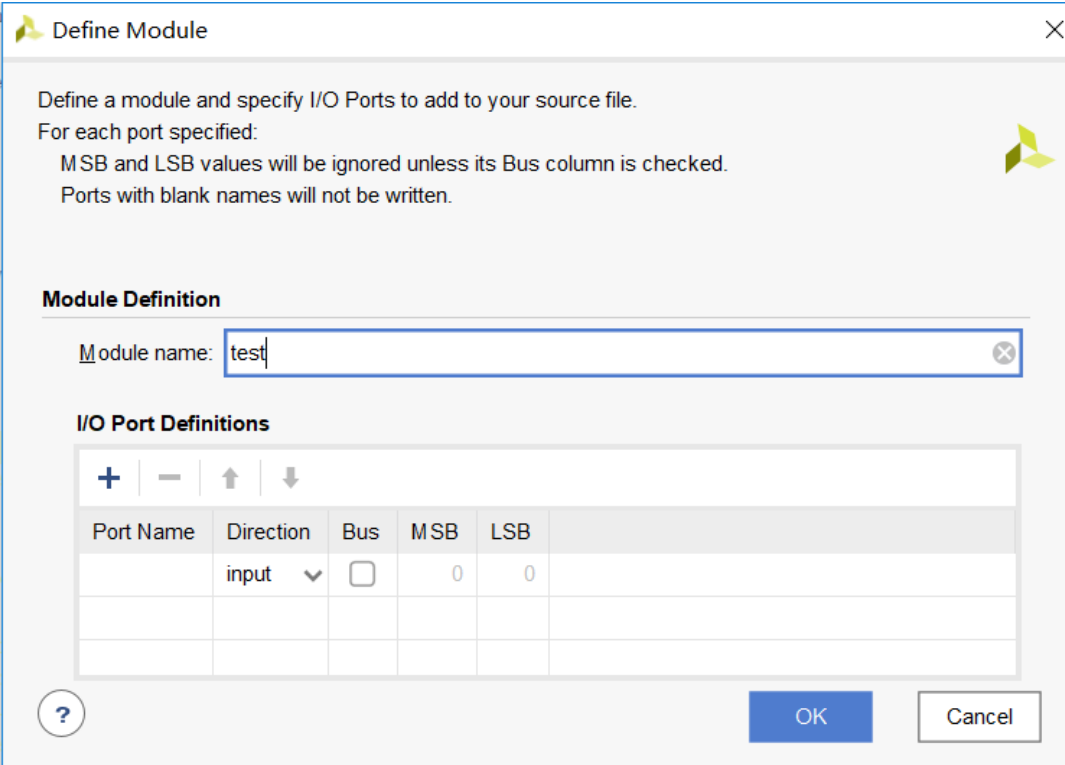
Write the File name, and click “OK”, File type selects “Verilog”



Click “**Finish**”



Click “**OK**”



Define a module and specify I/O Ports to add to your source file.  
For each port specified:  
MSB and LSB values will be ignored unless its Bus column is checked.  
Ports with blank names will not be written.

**Module Definition**

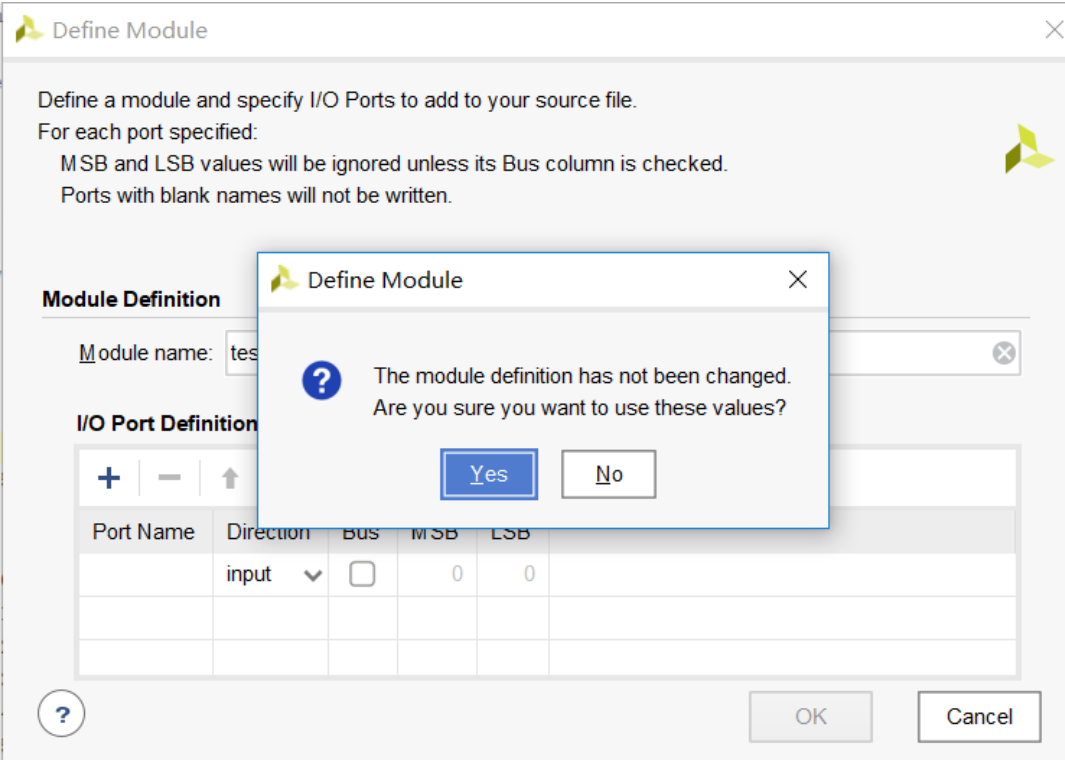
Module name:

**I/O Port Definitions**

| Port Name | Direction | Bus                      | MSB | LSB |
|-----------|-----------|--------------------------|-----|-----|
|           | input     | <input type="checkbox"/> | 0   | 0   |
|           |           |                          |     |     |
|           |           |                          |     |     |

Buttons: ? OK Cancel

Click “Yes”



Define a module and specify I/O Ports to add to your source file.  
For each port specified:  
MSB and LSB values will be ignored unless its Bus column is checked.  
Ports with blank names will not be written.

**Module Definition**

Module name:

**I/O Port Definition**

| Port Name | Direction | Bus                      | MSB | LSB |
|-----------|-----------|--------------------------|-----|-----|
|           | input     | <input type="checkbox"/> | 0   | 0   |
|           |           |                          |     |     |
|           |           |                          |     |     |

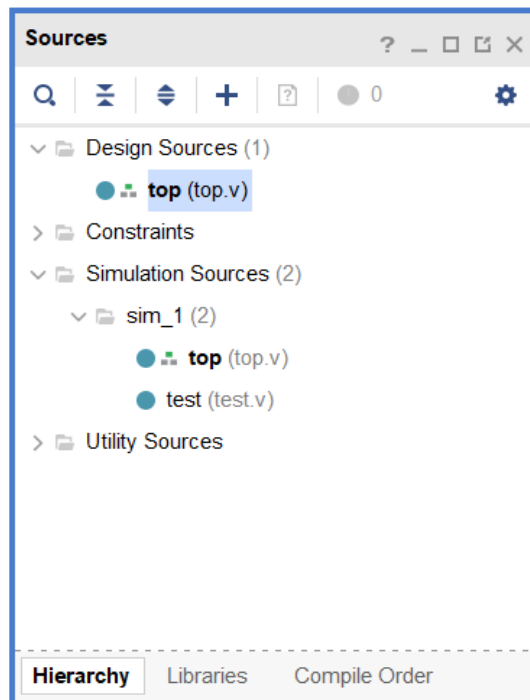
Buttons: ? OK Cancel

**Define Module**

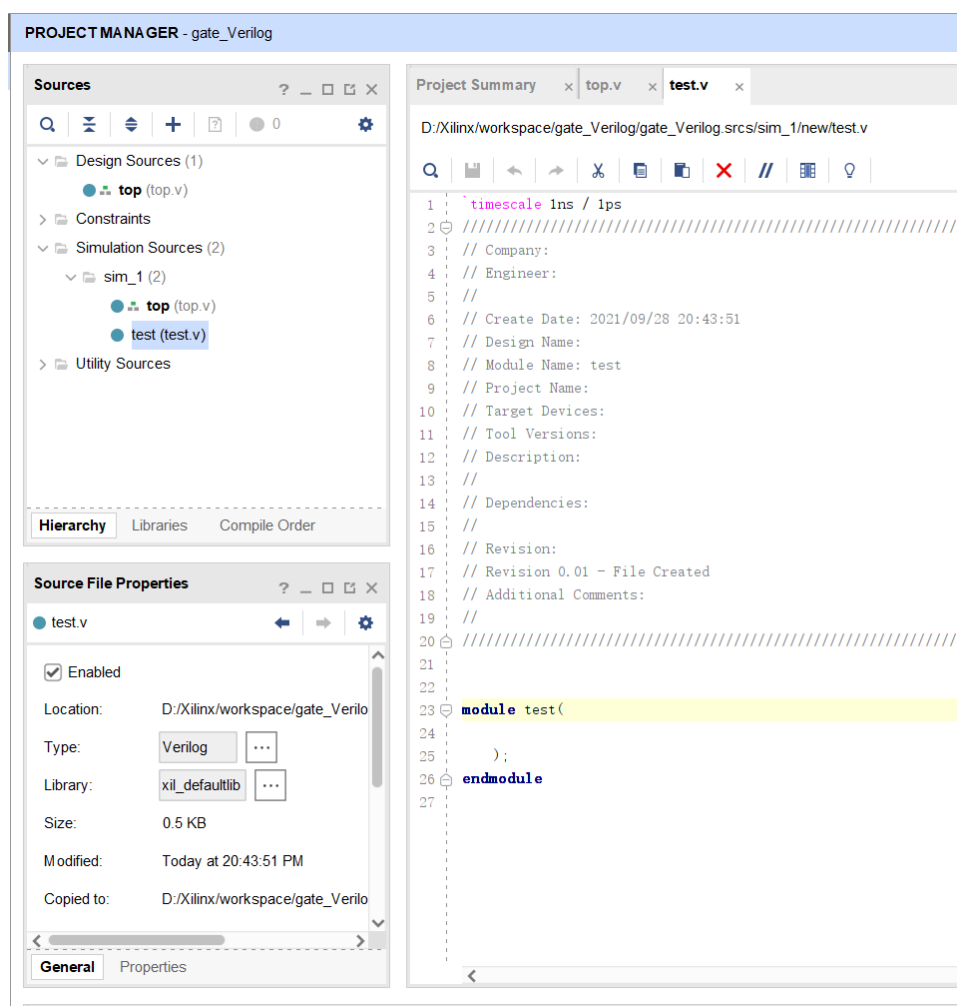
? The module definition has not been changed.  
Are you sure you want to use these values?

Buttons: Yes No

We have added test file under “**Simulation Sources**”



Open “test.v” file.



Then add test code as follows:

**test.v**

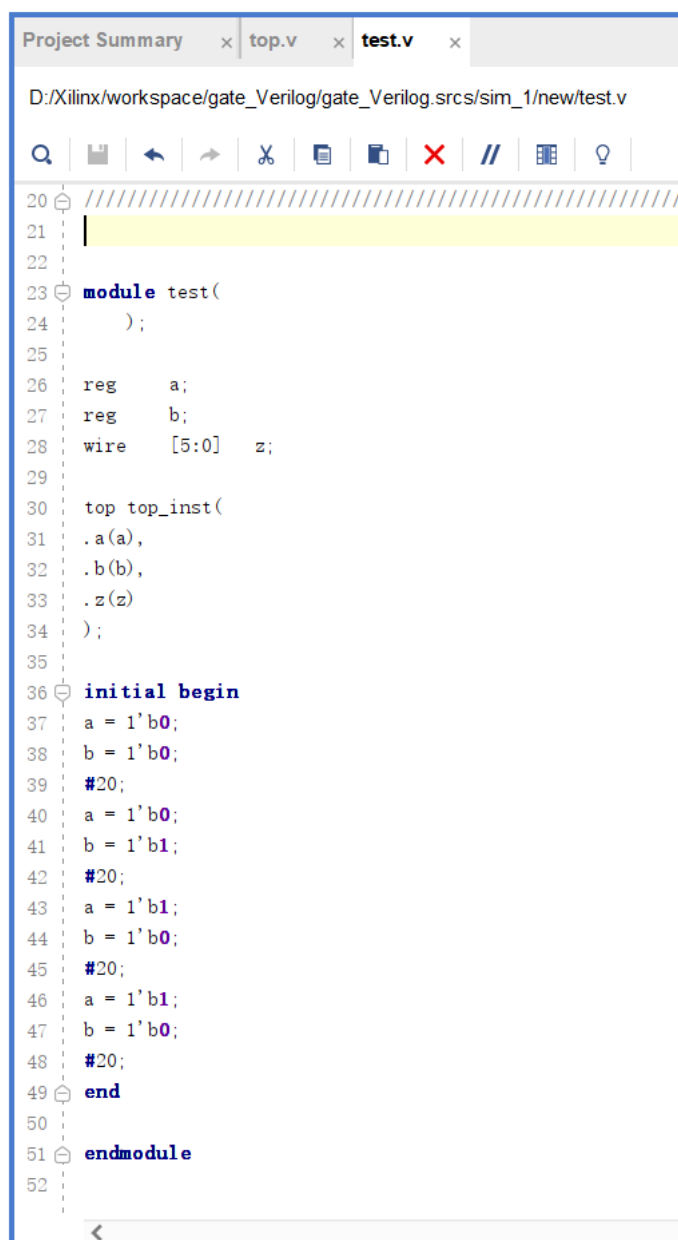
```
`timescale 1ns/1ps
module test(
    );

    reg      a;
    reg      b;
    wire      [5:0]  z;

    top top_inst(
        .a(a),
        .b(b),
        .z(z)
    );

    initial begin
        a = 1b0;
        b = 1b0;
        #20;
        a = 1b0;
        b = 1b1;
        #20;
        a = 1b1;
        b = 1b0;
        #20;
        a = 1b1;
        b = 1b 1;
        #20;
    end

endmodule
```



The screenshot shows a Verilog code editor with a tab labeled 'test.v'. The code is as follows:

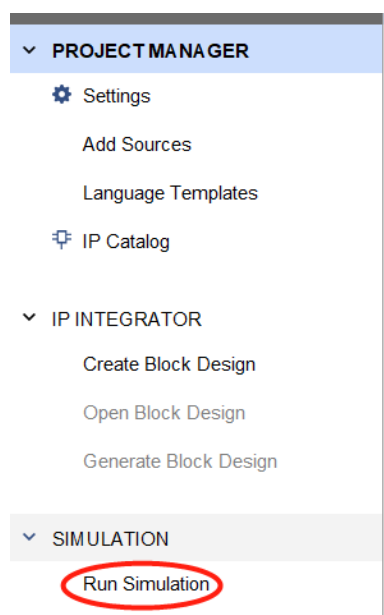
```
20 ///////////////////////////////////////////////////////////////////
21 |
22 |
23 module test(
24     );
25 |
26     reg    a;
27     reg    b;
28     wire   [5:0]  z;
29 |
30     top top_inst(
31         .a(a),
32         .b(b),
33         .z(z)
34     );
35 |
36     initial begin
37         a = 1'b0;
38         b = 1'b0;
39         #20;
40         a = 1'b0;
41         b = 1'b1;
42         #20;
43         a = 1'b1;
44         b = 1'b0;
45         #20;
46         a = 1'b1;
47         b = 1'b0;
48         #20;
49     end
50 |
51 endmodule
52 |
```

Click **save**

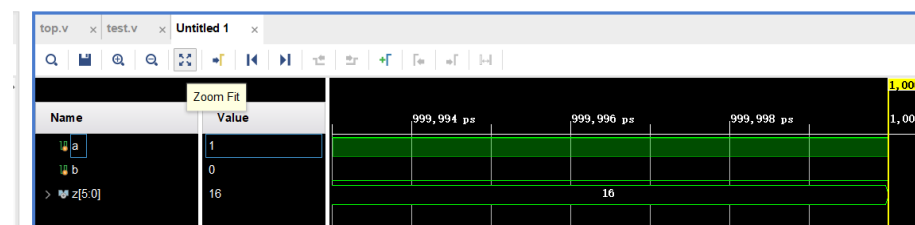


Click **“Run Simulation”**, select **“Run Behavioral Simulation”**





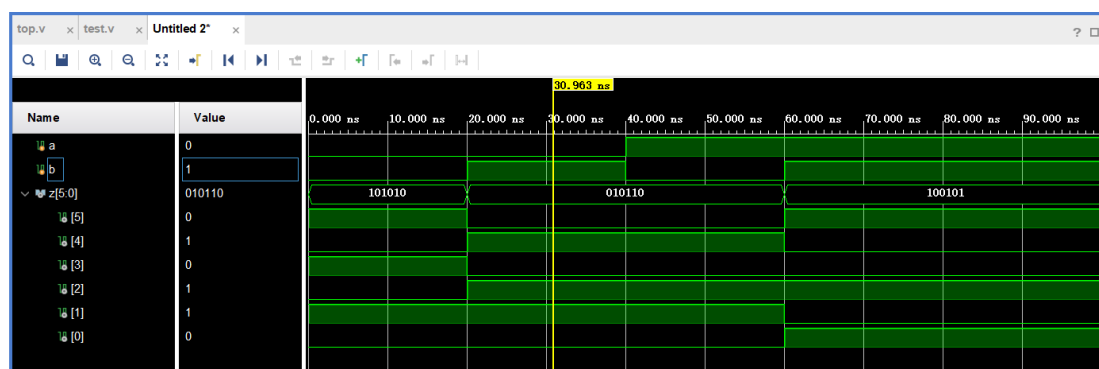
Click Zoom fit



right click Value of z[5:0], click “Radix”, click “Binary”, check results with truth table.

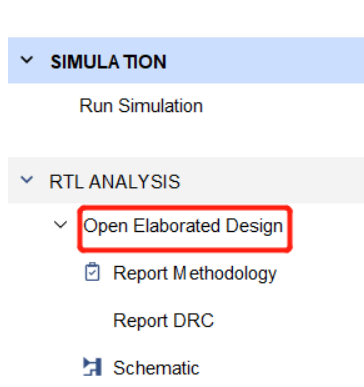
Truth Table

| a | b | Z[5:0] |
|---|---|--------|
| 0 | 0 | 101010 |
| 0 | 1 | 010110 |
| 1 | 0 | 010110 |
| 1 | 1 | 100101 |

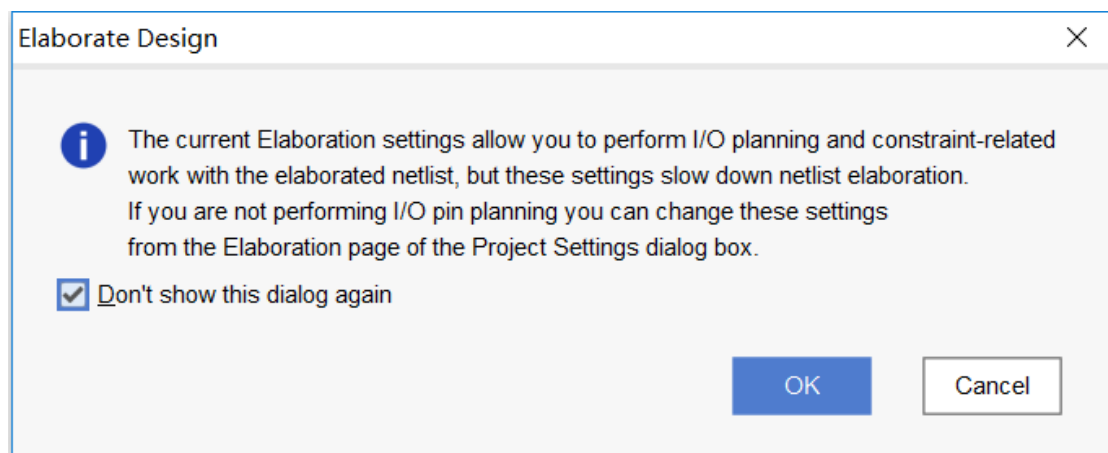


## Step4: RTL Analysis

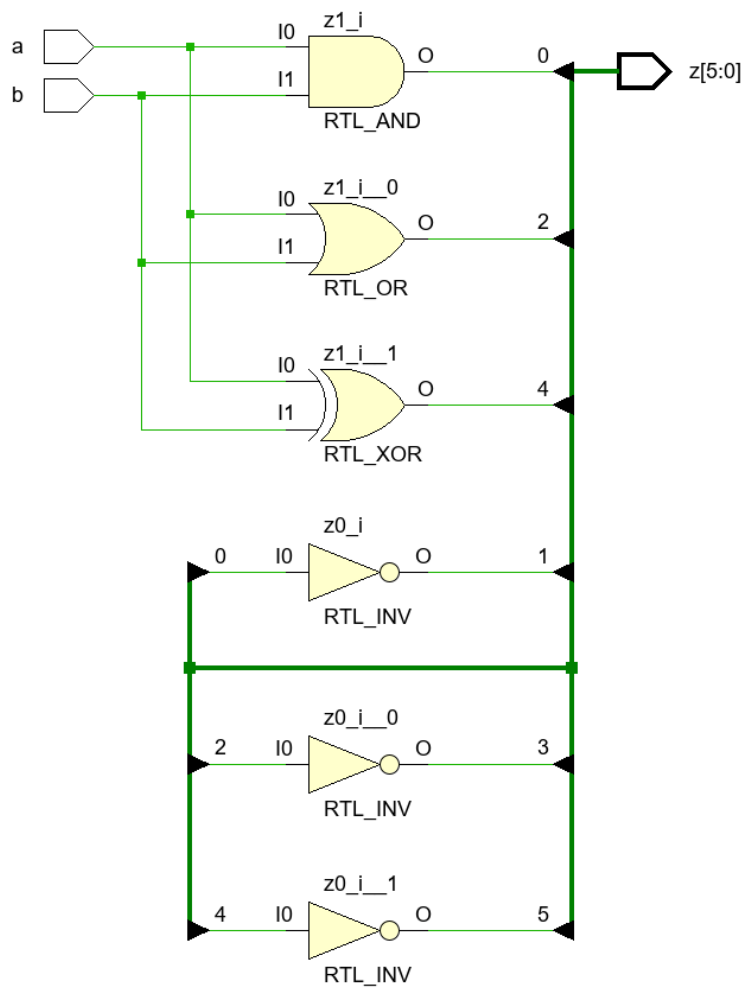
Click “Open Elaborated Design”



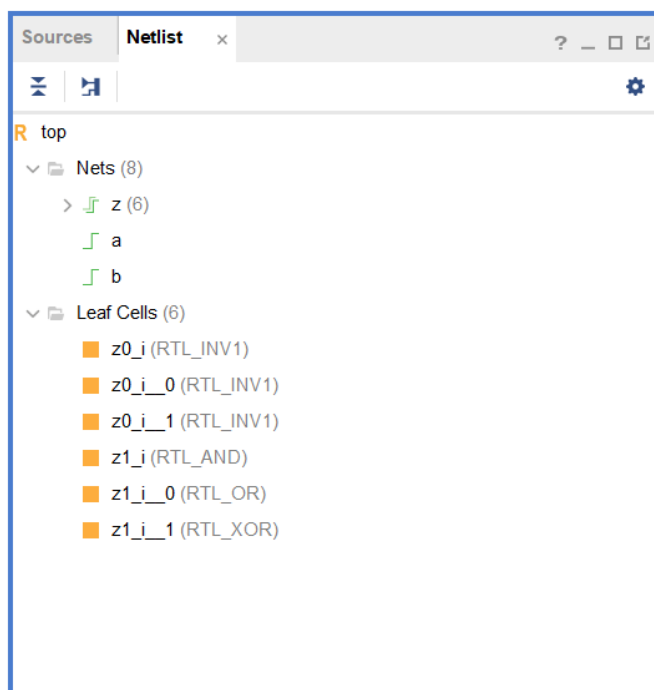
Click “OK”



Click “**Schematic**” after RTL process.

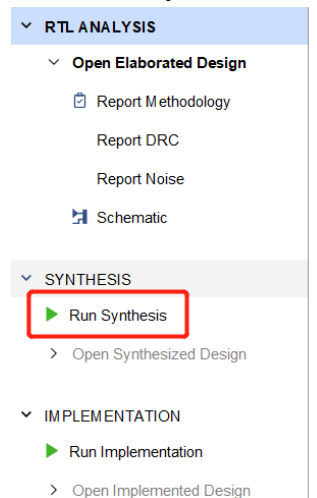


View “RTL Netlist”

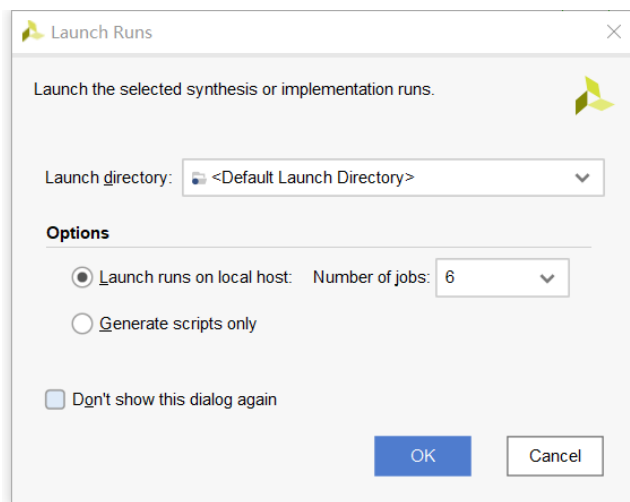


## Step5: Synthesis

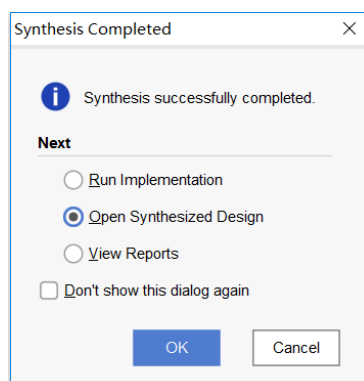
Click **“Run Synthesis”**



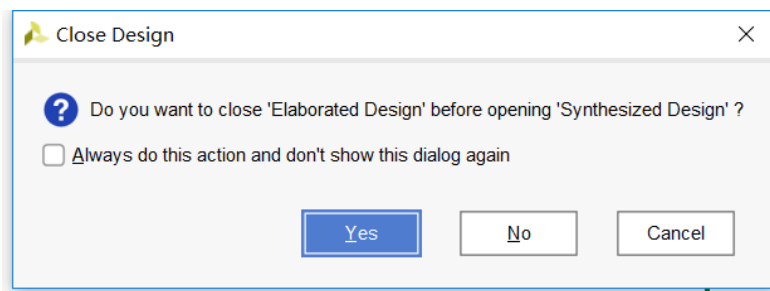
Click OK, to do synthesis.



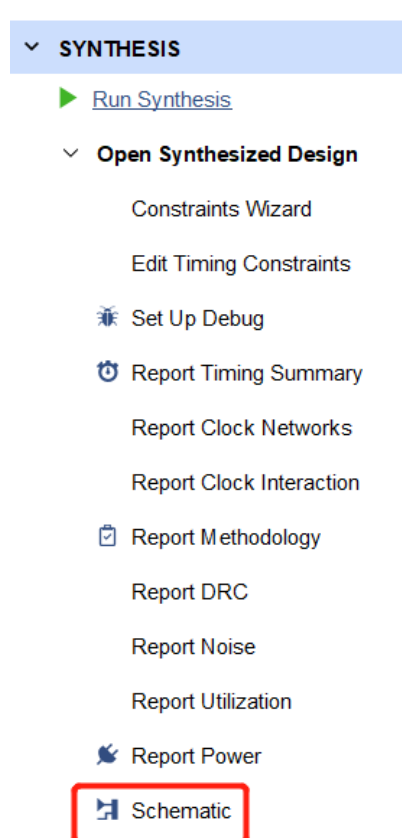
After Synthesis Completed, Select **“Open Synthesized Design”**, Click **“OK”**

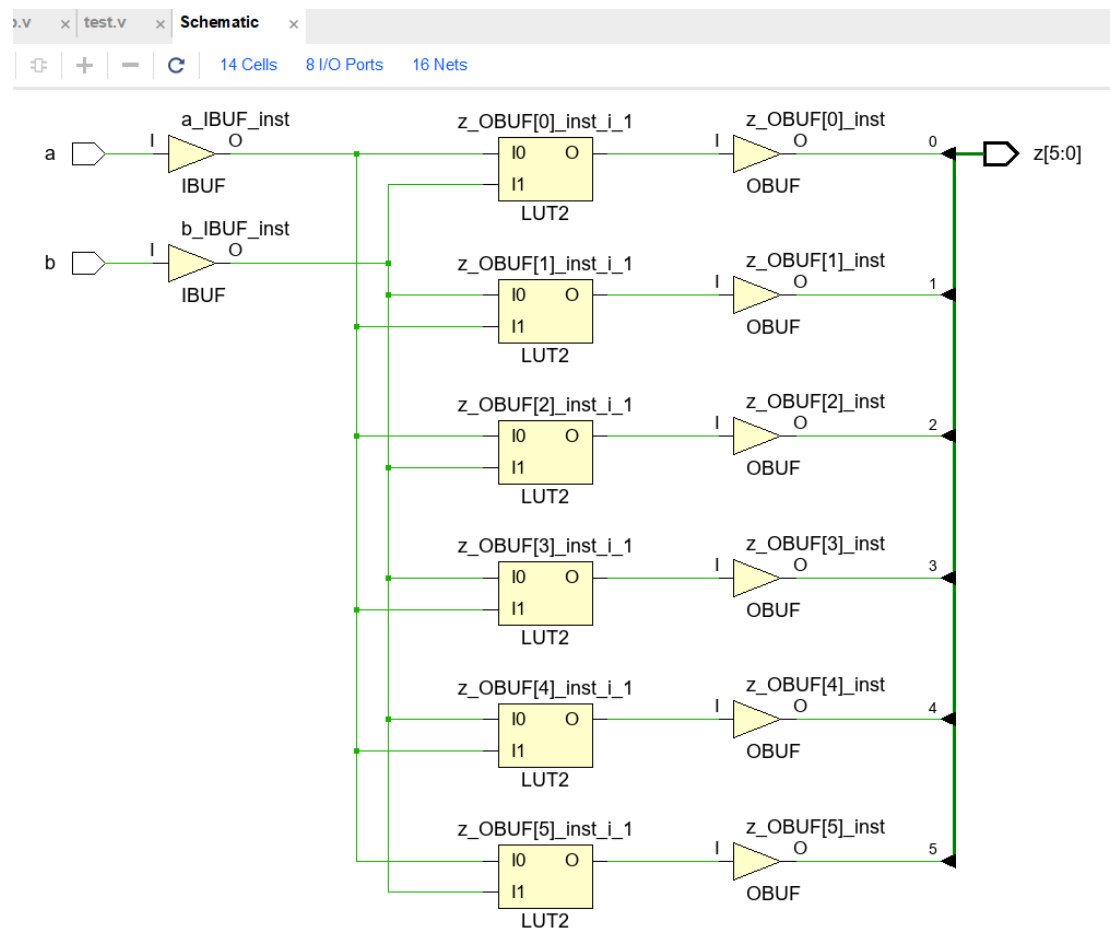


Click **“Yes”**

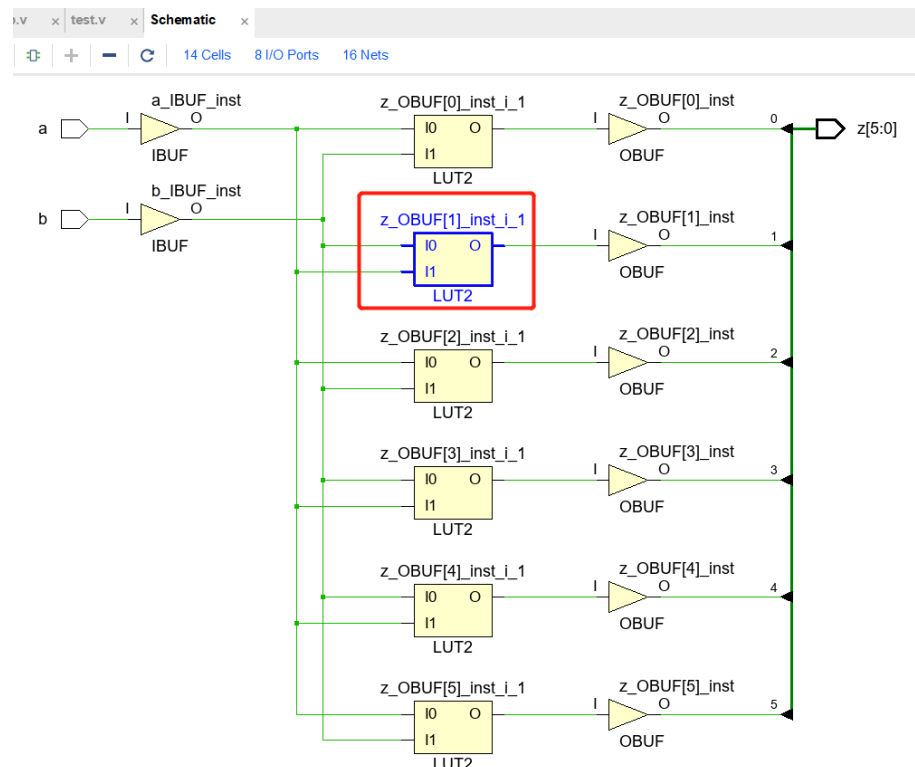


Click “**Schematic**”, and there are also many reports

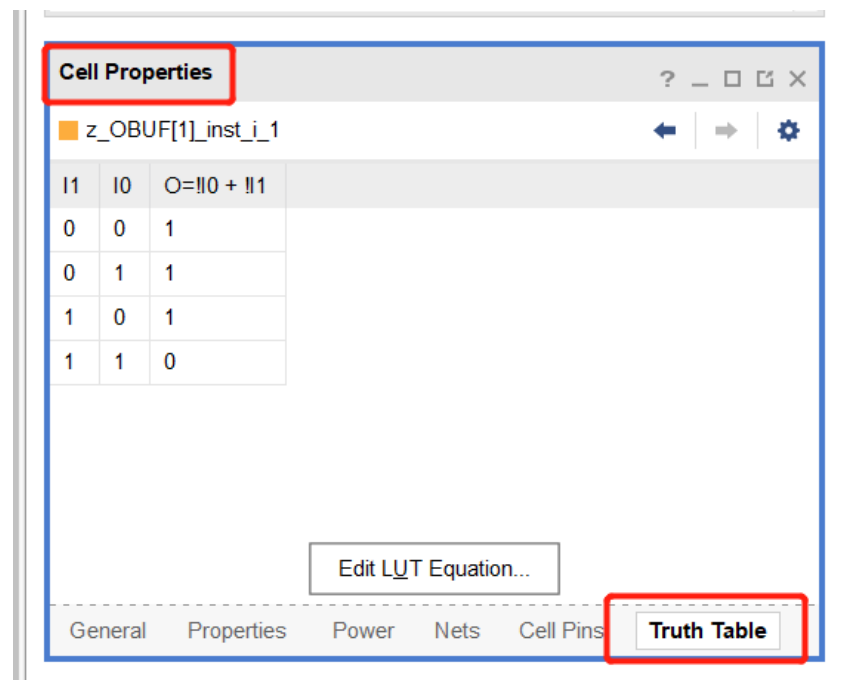




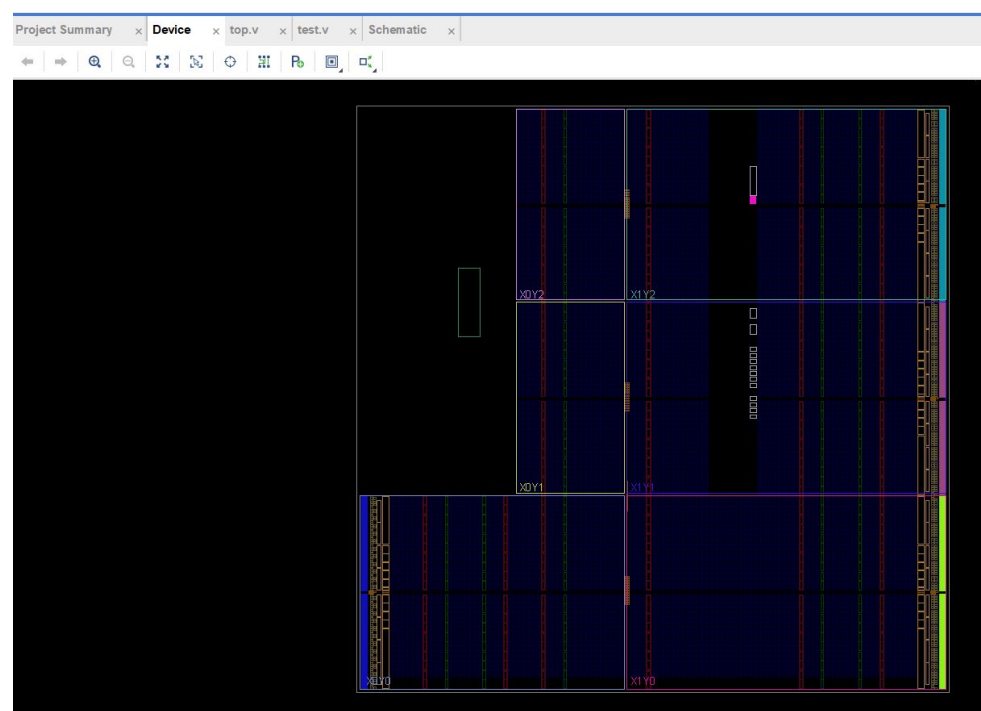
Double click one of LUTs.



Find Cell Properties ,Click “**Truth Table**”, we can check “**Truth Table**”

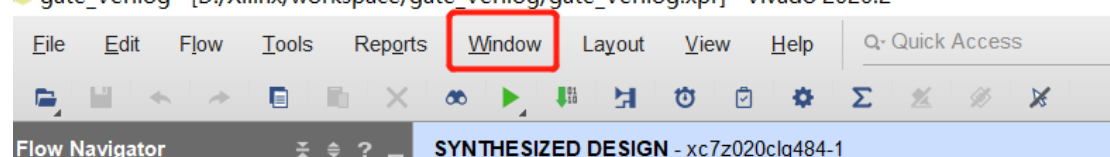


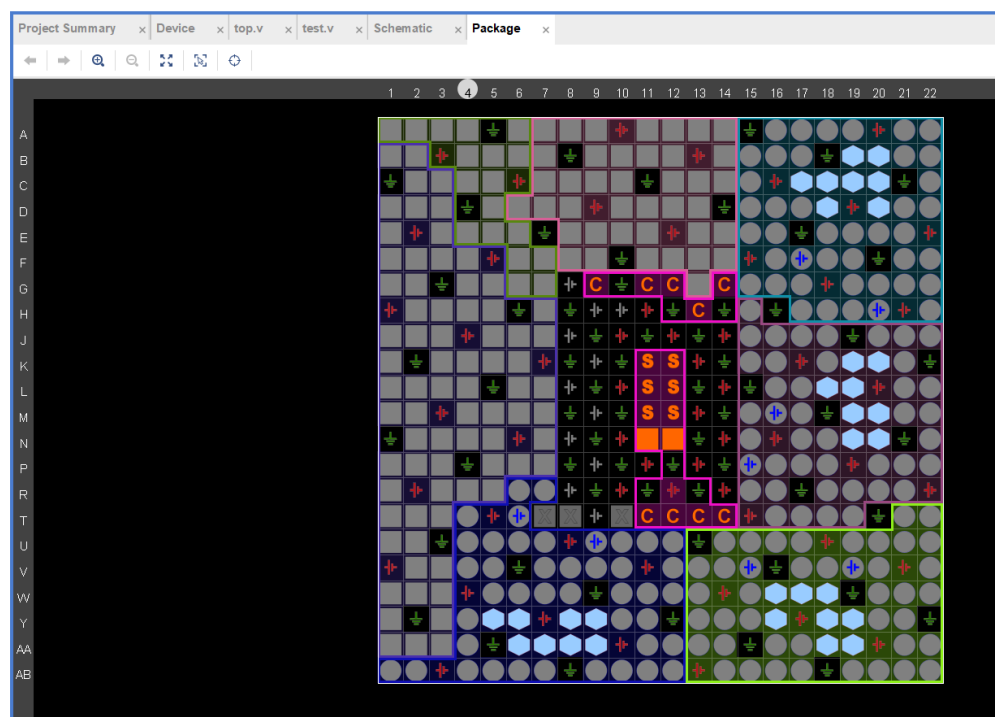
Also we can check “**Device**”



Also, we can click “**window**”, select “**Package**”

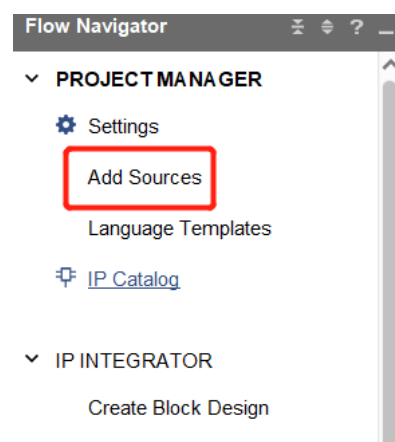
gate\_Verilog - [D:/Xilinx/workspace/gate\_Verilog/gate\_Verilog.xpr] - Vivado 2020.2





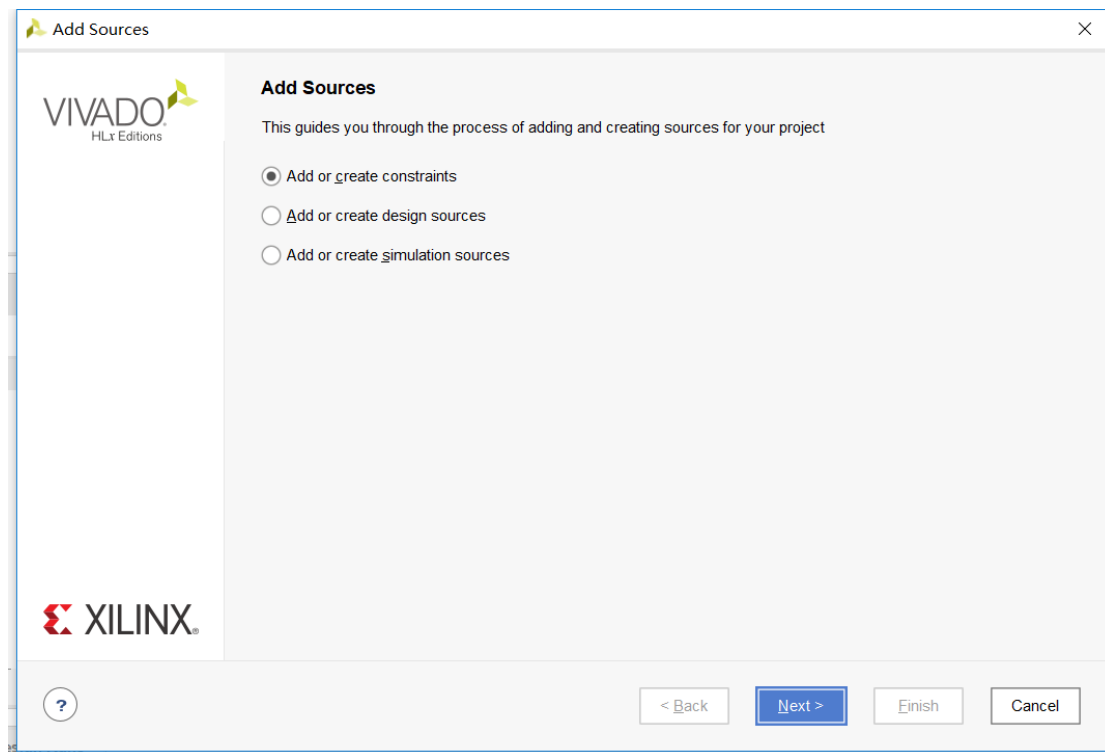
## Step6: Add constraints

Click “Add Sources”

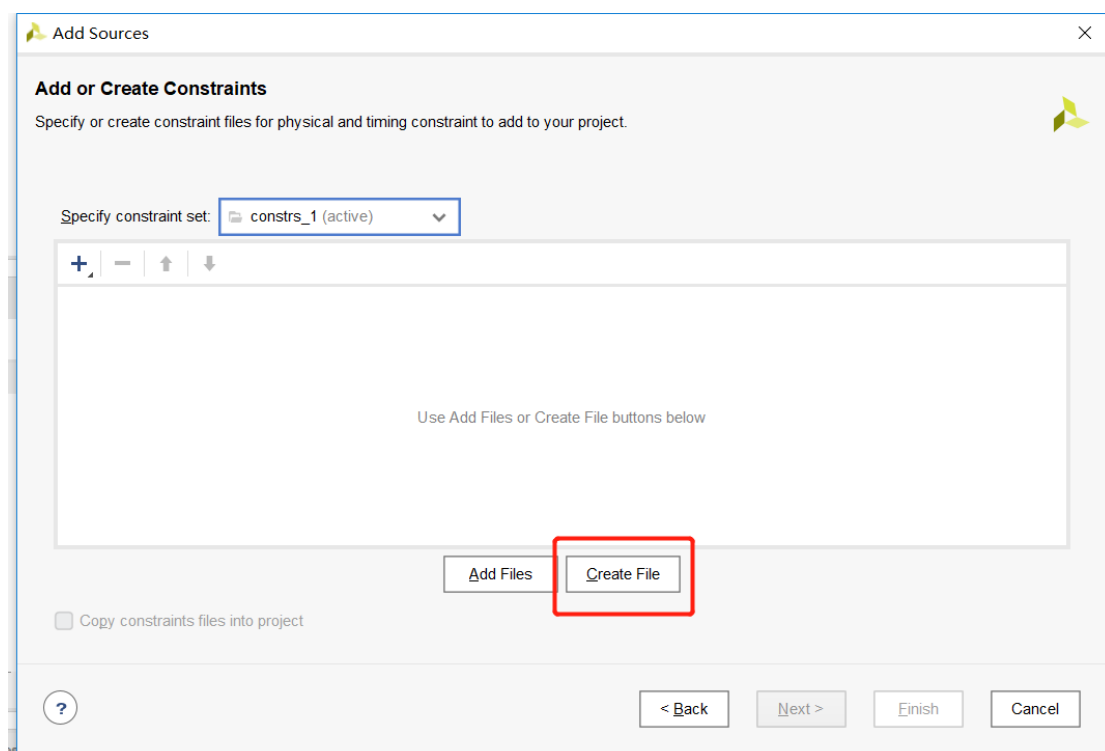


Select “Add or create constraints”, and click “Next”

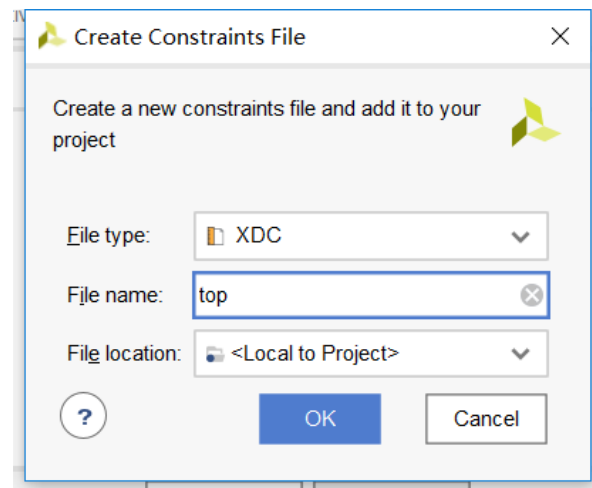




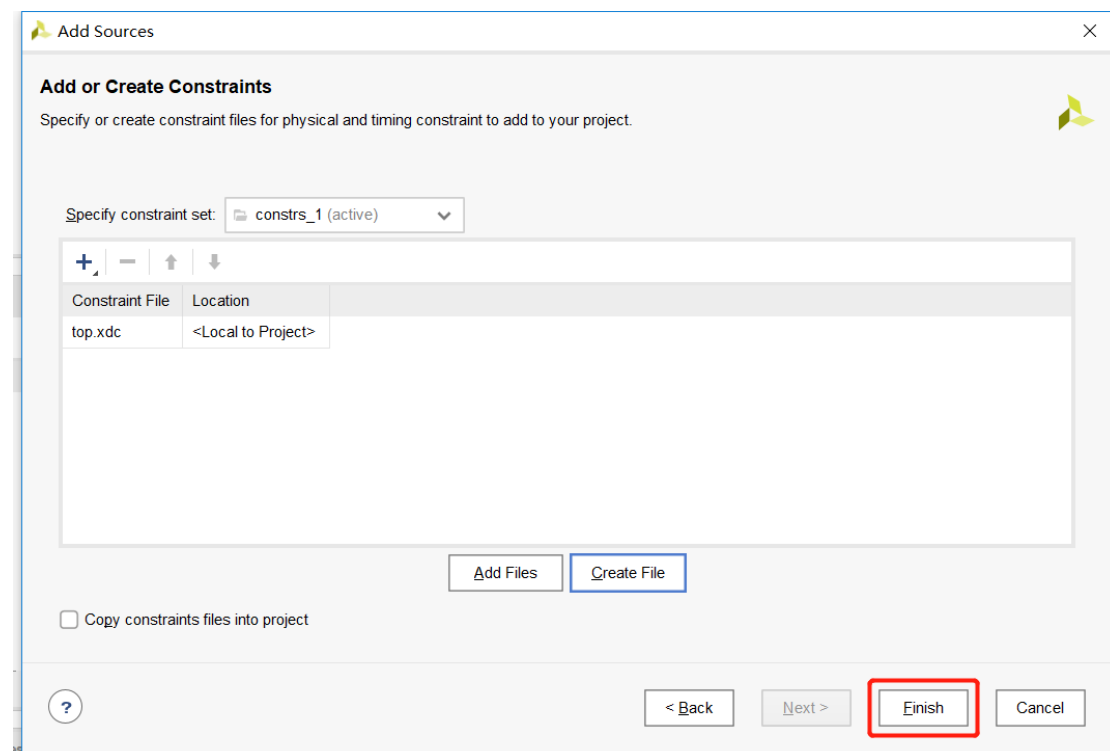
Click “**Create File**”



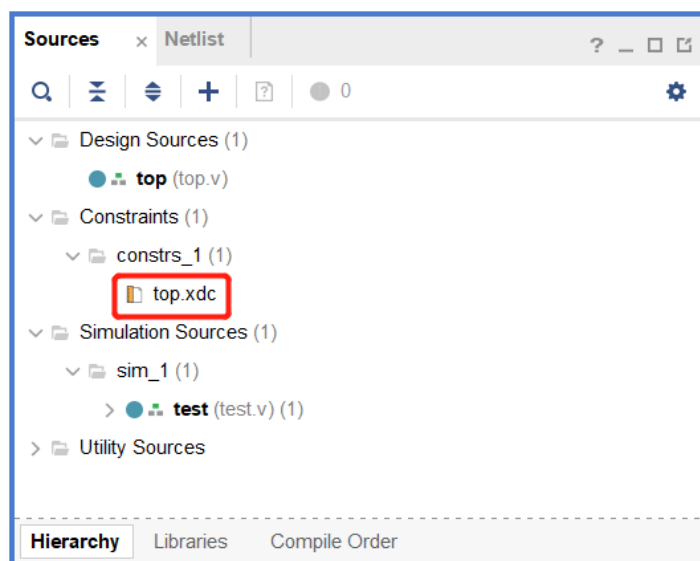
Write the File name, then click “**OK**”



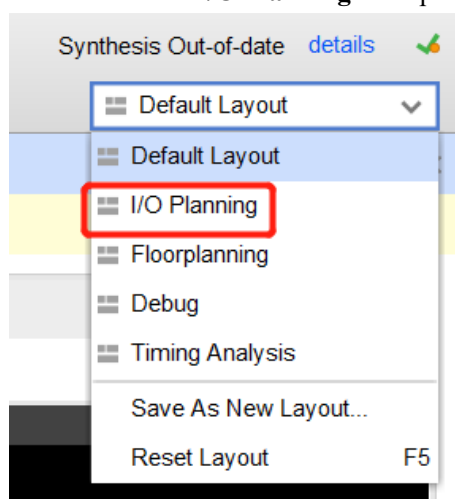
Click “**Finish**”



Then we have added Constraints file: **top.xdc**



Then we select “I/O Planning” at top right corner.



Next, we will assign Pins,

Tcl ConsoleMessagesLogReportsDesign RunsPackage PinsI/O Ports x

Q

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| Name             | Direction | Board Part Pin | Board Part Interface | Neg Diff Pair | Package Pin | Fixed                    | Bank | I/O Std            | Vcco  | Vref | Drive Strength | Slew Type | Pull Type | Off-Chip Termination | IN_TERM |
|------------------|-----------|----------------|----------------------|---------------|-------------|--------------------------|------|--------------------|-------|------|----------------|-----------|-----------|----------------------|---------|
| All ports (8)    |           |                |                      |               |             |                          |      |                    |       |      |                |           |           |                      |         |
| z[5]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| z[4]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| z[3]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| z[2]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| z[1]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| z[0]             | OUT       |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      | 12             |           | NONE      | FP_VTT_50            |         |
| Scalar ports (2) |           |                |                      |               |             |                          |      |                    |       |      |                |           |           |                      |         |
| a                | IN        |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      |                |           | NONE      | NONE                 |         |
| b                | IN        |                |                      |               |             | <input type="checkbox"/> |      | default (LVCMOS18) | 1.800 |      |                |           | NONE      | NONE                 |         |

By Zedborad’s Hardware User’s guide

**Table 13 - DIP Switch Connections**

| Signal Name | Zynq pin |
|-------------|----------|
| SW0         | F22      |
| SW1         | G22      |
| SW2         | H22      |
| SW3         | F21      |
| SW4         | H19      |
| SW5         | H18      |
| SW6         | H17      |
| SW7         | M15      |

**Table 14 - LED Connections**

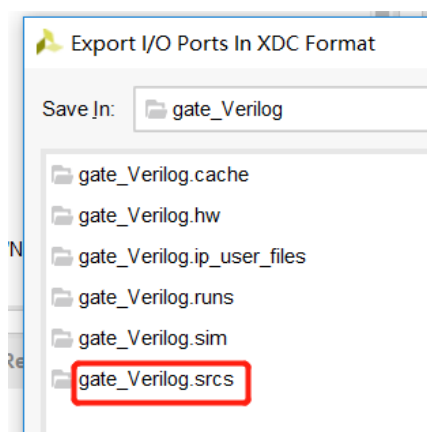
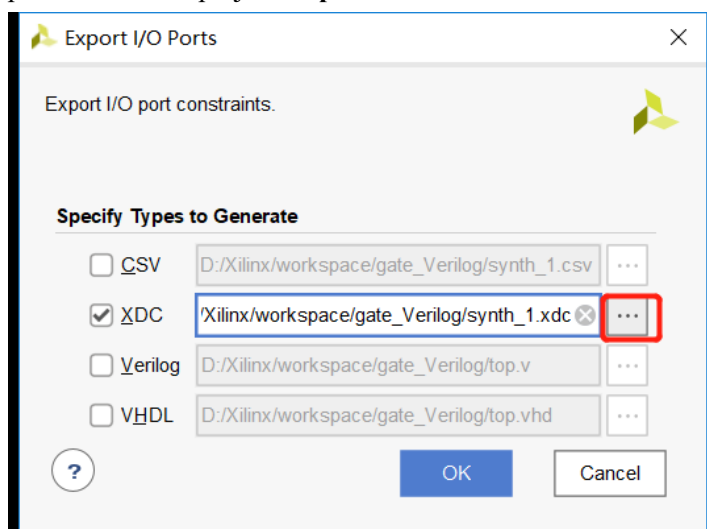
| Signal Name | Subsection | Zynq pin  |
|-------------|------------|-----------|
| LD0         | PL         | T22       |
| LD1         | PL         | T21       |
| LD2         | PL         | U22       |
| LD3         | PL         | U21       |
| LD4         | PL         | V22       |
| LD5         | PL         | W22       |
| LD6         | PL         | U19       |
| LD7         | PL         | U14       |
| LD9         | PS         | D5 (MIO7) |

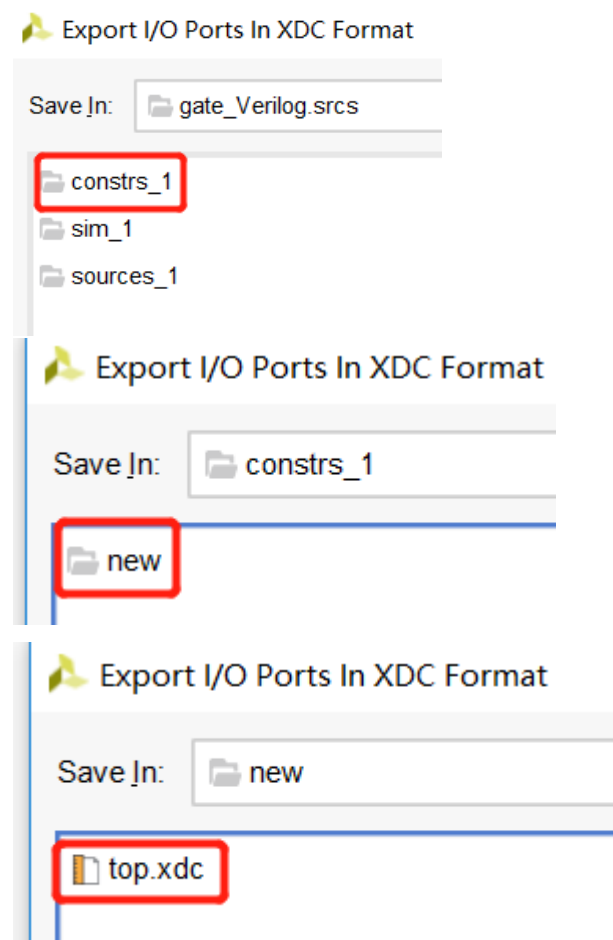
**Assign ports**

| Tcl Console      | Messages  | Log            | Reports              | Design Runs   | Package Pins | I/O Ports                           | x    |            |       |     |
|------------------|-----------|----------------|----------------------|---------------|--------------|-------------------------------------|------|------------|-------|-----|
|                  |           |                |                      |               |              |                                     |      |            |       |     |
| Name             | Direction | Board Part Pin | Board Part Interface | Neg Diff Pair | Package Pin  | Fixed                               | Bank | I/O Std    | Vcco  | Vre |
| All ports (8)    |           |                |                      |               |              |                                     |      |            |       |     |
| z (6)            | OUT       |                |                      |               |              | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[5]             | OUT       |                |                      |               | W22          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[4]             | OUT       |                |                      |               | V22          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[3]             | OUT       |                |                      |               | U21          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[2]             | OUT       |                |                      |               | U22          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[1]             | OUT       |                |                      |               | T21          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| z[0]             | OUT       |                |                      |               | T22          | <input checked="" type="checkbox"/> | 33   | LVC MOS33* | 3.300 |     |
| Scalar ports (2) |           |                |                      |               |              |                                     |      |            |       |     |
| a                | IN        |                |                      |               | F22          | <input checked="" type="checkbox"/> | 35   | LVC MOS33* | 3.300 |     |
| b                | IN        |                |                      |               | G22          | <input checked="" type="checkbox"/> | 35   | LVC MOS33* | 3.300 |     |

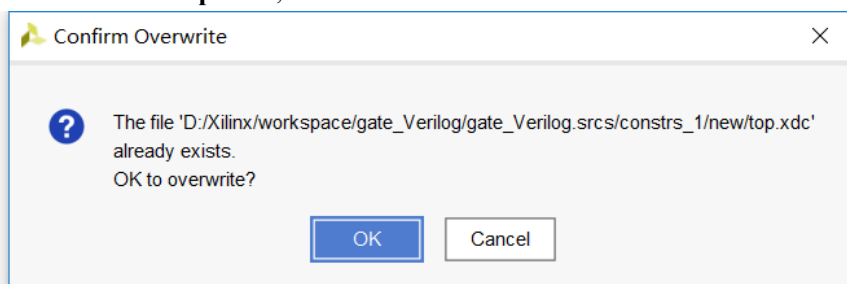
| port | pin | I/O standard |
|------|-----|--------------|
| a    | F22 | LVC MOS33    |
| b    | G22 | LVC MOS33    |
| z[0] | T22 | LVC MOS33    |
| z[1] | T21 | LVC MOS33    |
| z[2] | U22 | LVC MOS33    |
| z[3] | U21 | LVC MOS33    |
| z[4] | V22 | LVC MOS33    |
| z[5] | W22 | LVC MOS33    |

Right click, select “**Export I/O Ports**” in the figure above. **Select XDC**, don’t select CSV, and path points to current project “**top.xdc**”

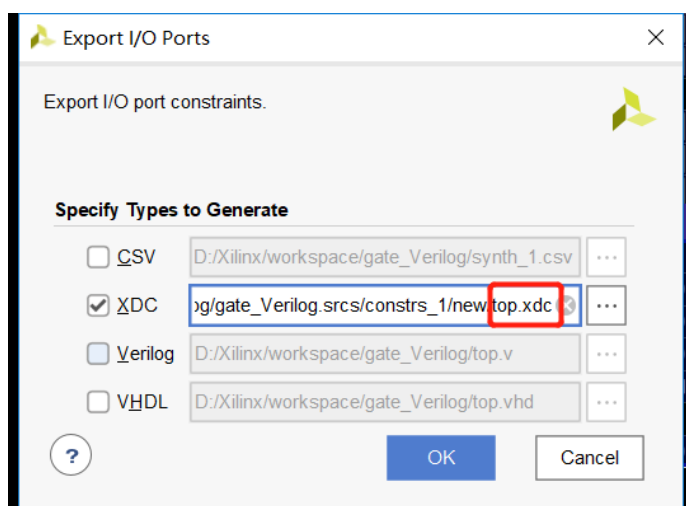




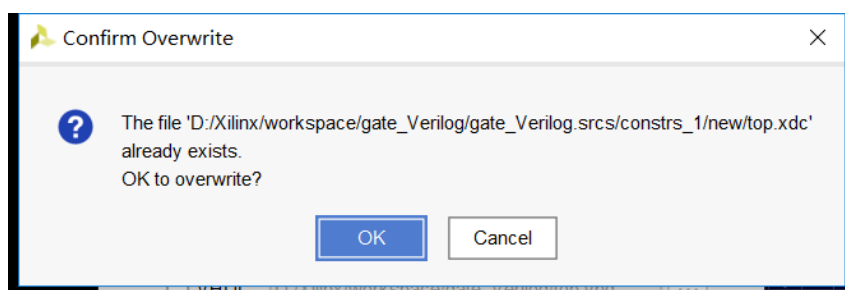
Double click **“top.xdc”**, then click **“Ok”**



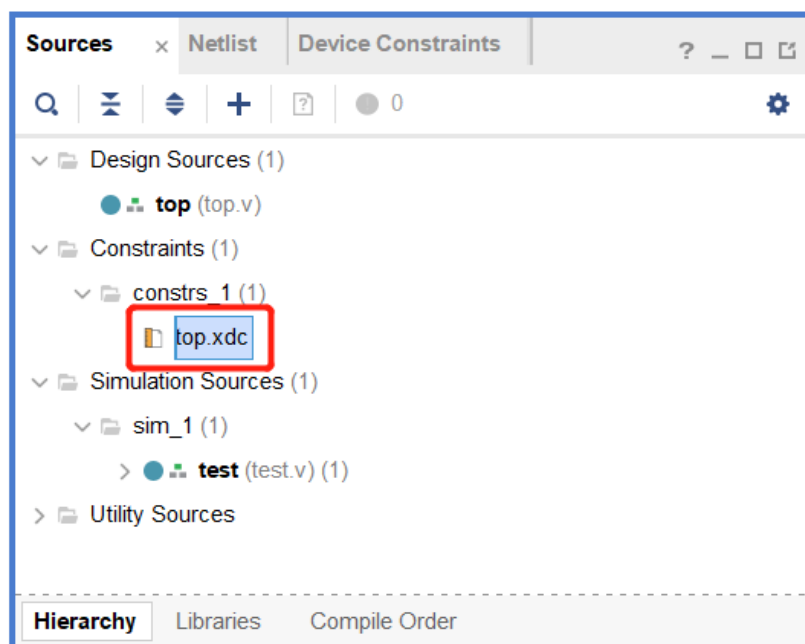
Click **“OK”**



Click "OK"



Find and open "top.xdc".and we can check constraints code.

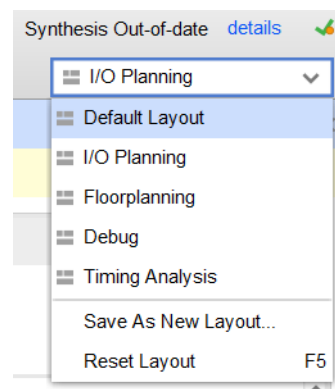


```

1  set_property PACKAGE_PIN F22 [get_ports a]
2  set_property PACKAGE_PIN G22 [get_ports b]
3  set_property PACKAGE_PIN T22 [get_ports {z[0]}]
4  set_property PACKAGE_PIN T21 [get_ports {z[1]}]
5  set_property PACKAGE_PIN U22 [get_ports {z[2]}]
6  set_property PACKAGE_PIN U21 [get_ports {z[3]}]
7  set_property PACKAGE_PIN V22 [get_ports {z[4]}]
8  set_property PACKAGE_PIN W22 [get_ports {z[5]}]
9  set_property DIRECTION OUT [get_ports {z[5]}]
10 set_property IOSTANDARD LVCMOS33 [get_ports {z[5]}]
11 set_property DRIVE 12 [get_ports {z[5]}]
12 set_property SLEW SLOW [get_ports {z[5]}]
13 set_property DIRECTION OUT [get_ports {z[4]}]
14 set_property IOSTANDARD LVCMOS33 [get_ports {z[4]}]
15 set_property DRIVE 12 [get_ports {z[4]}]
16 set_property SLEW SLOW [get_ports {z[4]}]
17 set_property DIRECTION OUT [get_ports {z[3]}]
18 set_property IOSTANDARD LVCMOS33 [get_ports {z[3]}]
19 set_property DRIVE 12 [get_ports {z[3]}]
20 set_property SLEW SLOW [get_ports {z[3]}]
21 set_property DIRECTION OUT [get_ports {z[2]}]
22 set_property IOSTANDARD LVCMOS33 [get_ports {z[2]}]
23 set_property DRIVE 12 [get_ports {z[2]}]
24 set_property SLEW SLOW [get_ports {z[2]}]
25 set_property DIRECTION OUT [get_ports {z[1]}]
26 set_property IOSTANDARD LVCMOS33 [get_ports {z[1]}]
27 set_property DRIVE 12 [get_ports {z[1]}]
28 set_property SLEW SLOW [get_ports {z[1]}]
29 set_property DIRECTION OUT [get_ports {z[0]}]
30 set_property IOSTANDARD LVCMOS33 [get_ports {z[0]}]
31 set_property DRIVE 12 [get_ports {z[0]}]
32 set_property SLEW SLOW [get_ports {z[0]}]
33 set_property DIRECTION IN [get_ports a]
34 set_property IOSTANDARD LVCMOS33 [get_ports a]
35 set_property DIRECTION IN [get_ports b]
36 set_property IOSTANDARD LVCMOS33 [get_ports b]
37 #revert back to original instance
38 current_instance -quiet

```

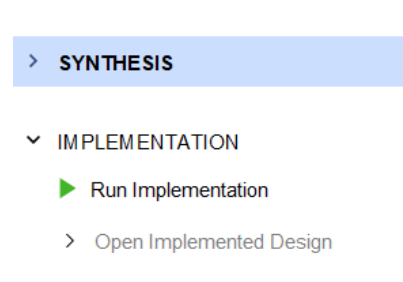
Return to “Default Layout” at right top corner.



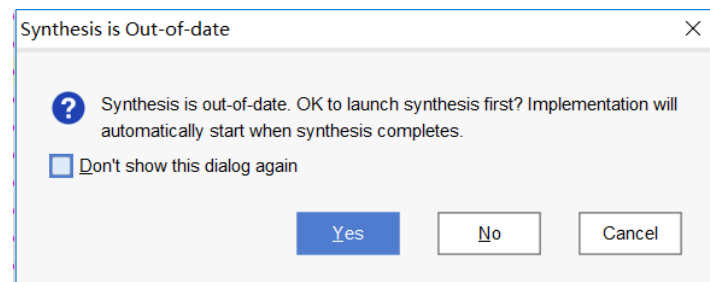


## Step7: Implementation

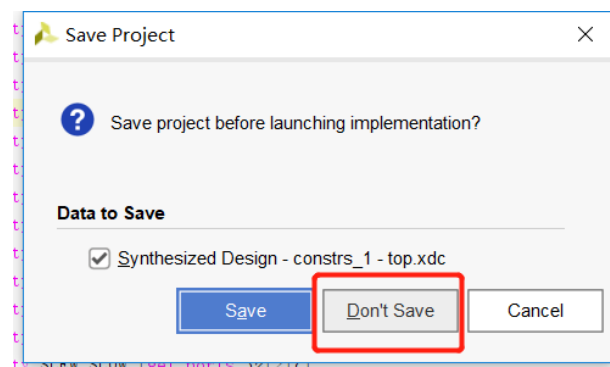
Click “Run Implementation”



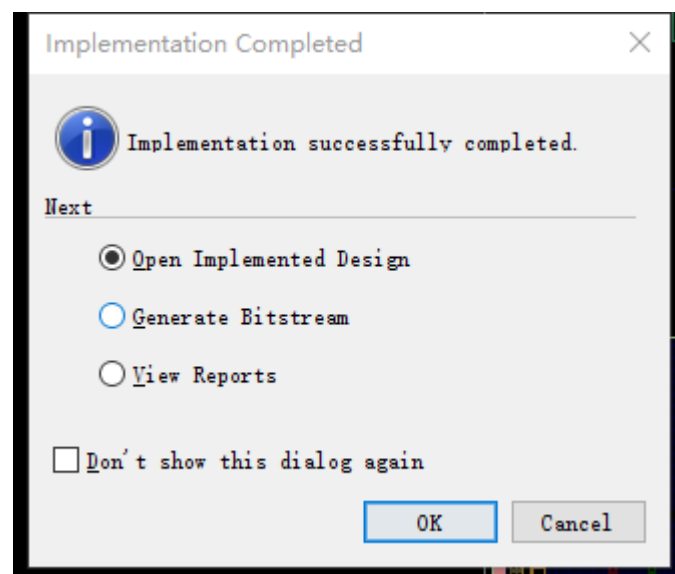
Click “Yes”



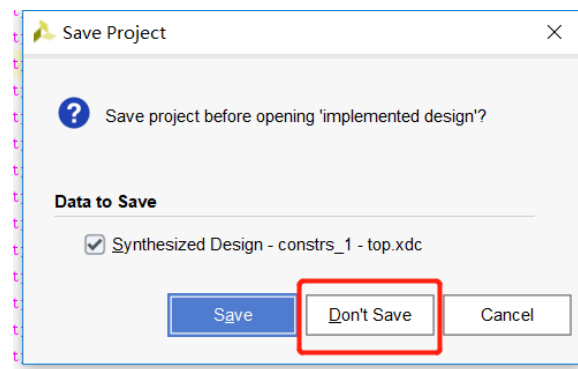
Click “Don't save”



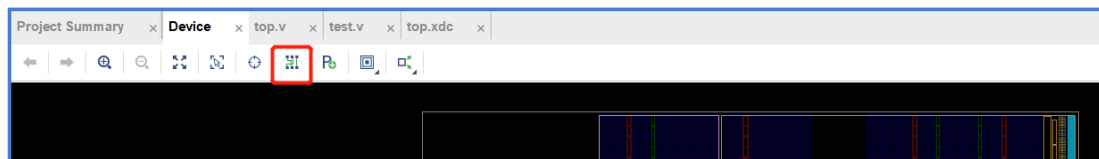
After Implementation, Click “OK” and open Implemented Design



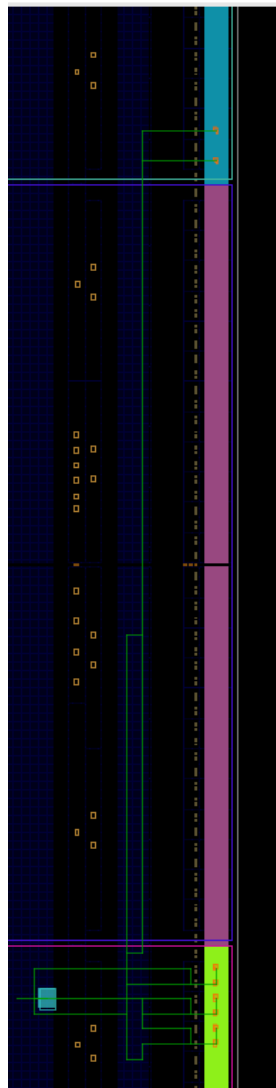
Click **“Don’t save”**

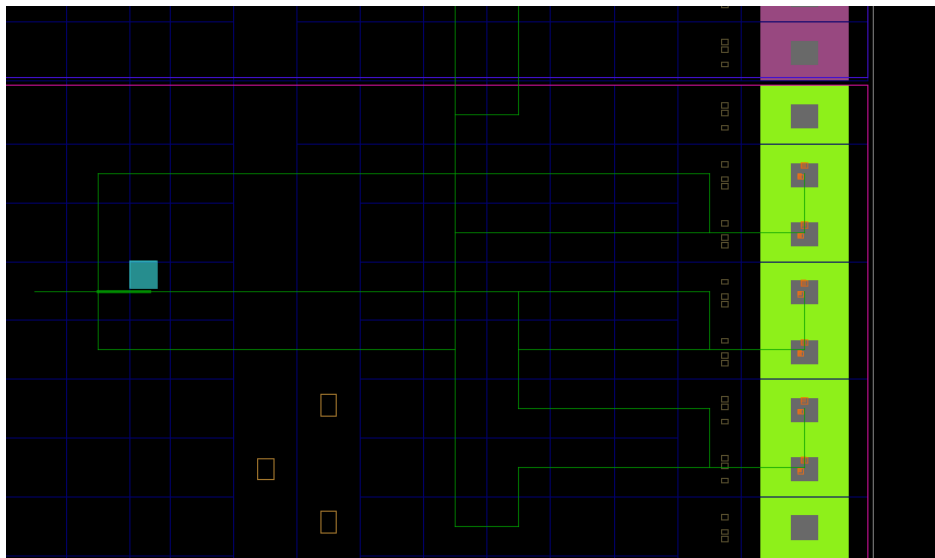


Click **“Routing Resources”**



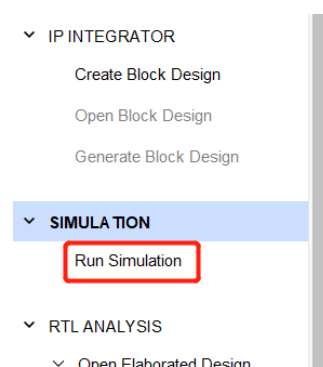
Click  under Device graph, adjust the suitable location. check routing



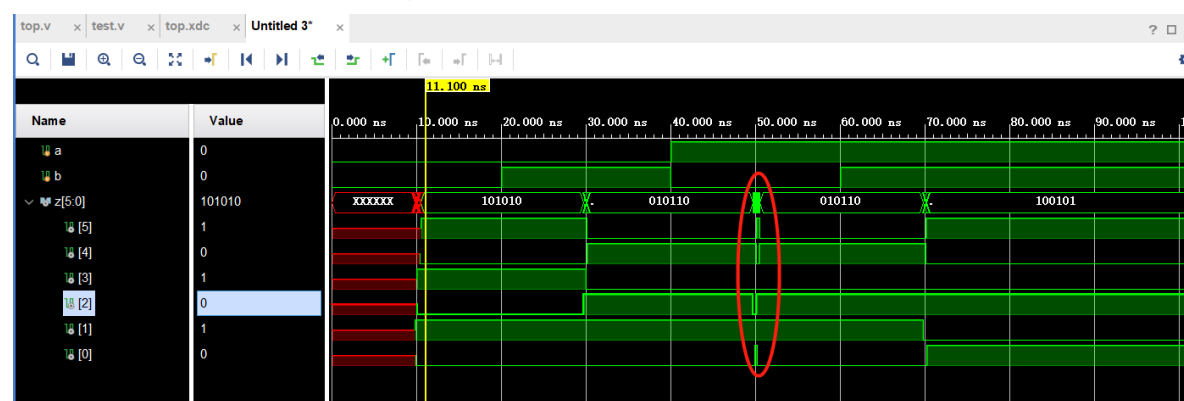


### Step8: Run Post-Implementation Timing Simulation

Click “Run Simulation”, select “Run Post-Implementation Timing Simulation”, Then select Don’t save.



We can see that there is a little delay from simulation.



### Step9: Generate Bitstream

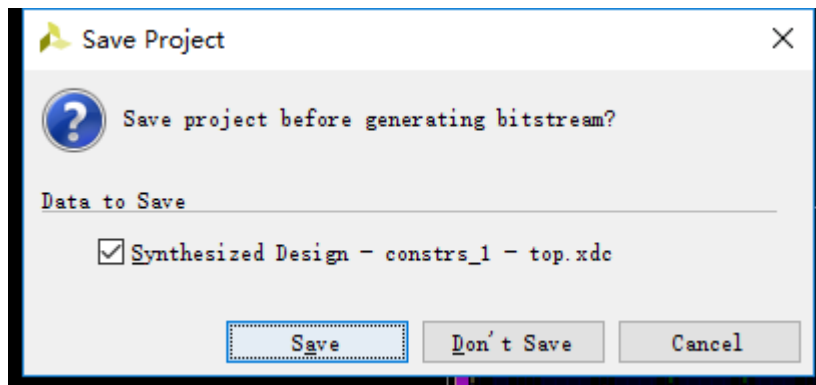
Click “Generate Bitstream”

## ▼ PROGRAM AND DEBUG

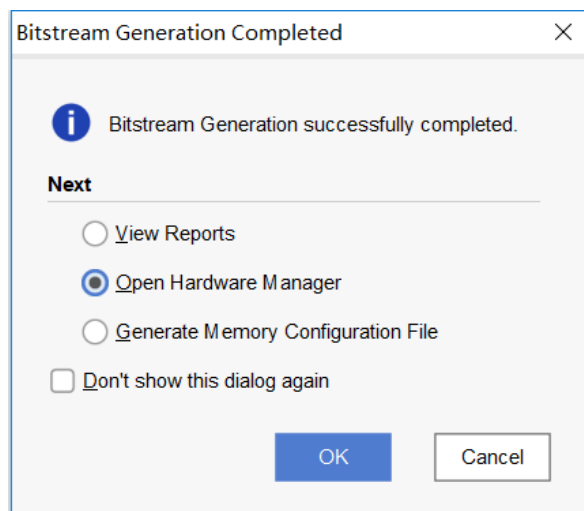


&gt; Open Hardware Manager

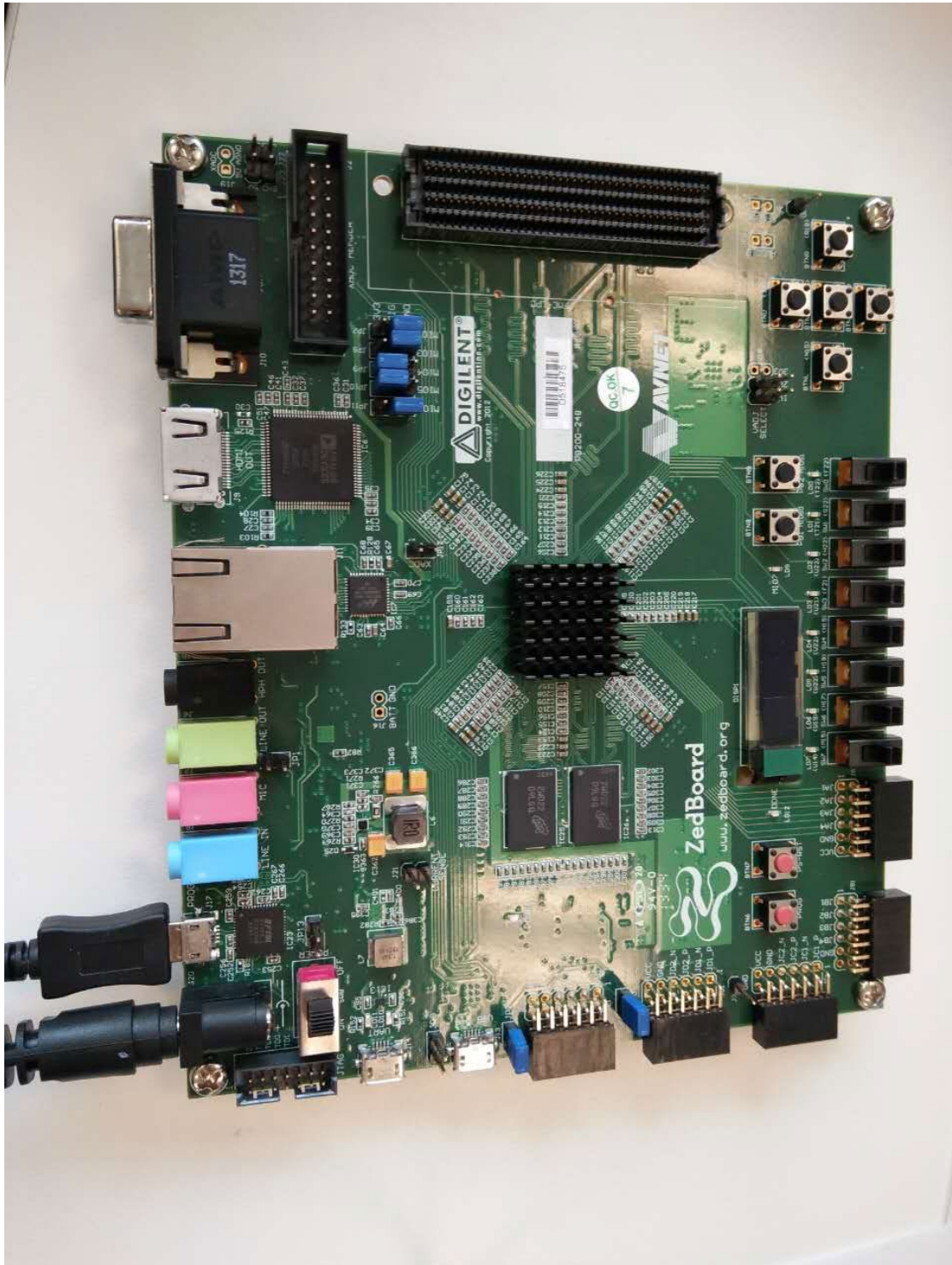
Click “Don’t Save”



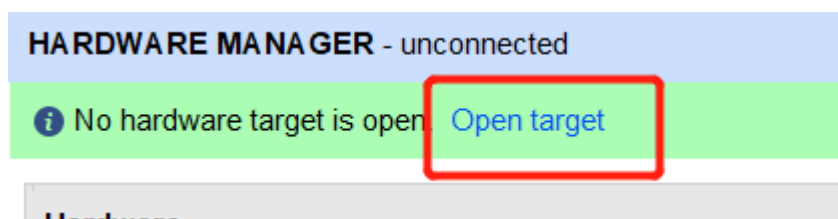
Select “Open Hardware Manager”, and click “OK”

**Step10: Connect Zedboard and Configure FPGA,**

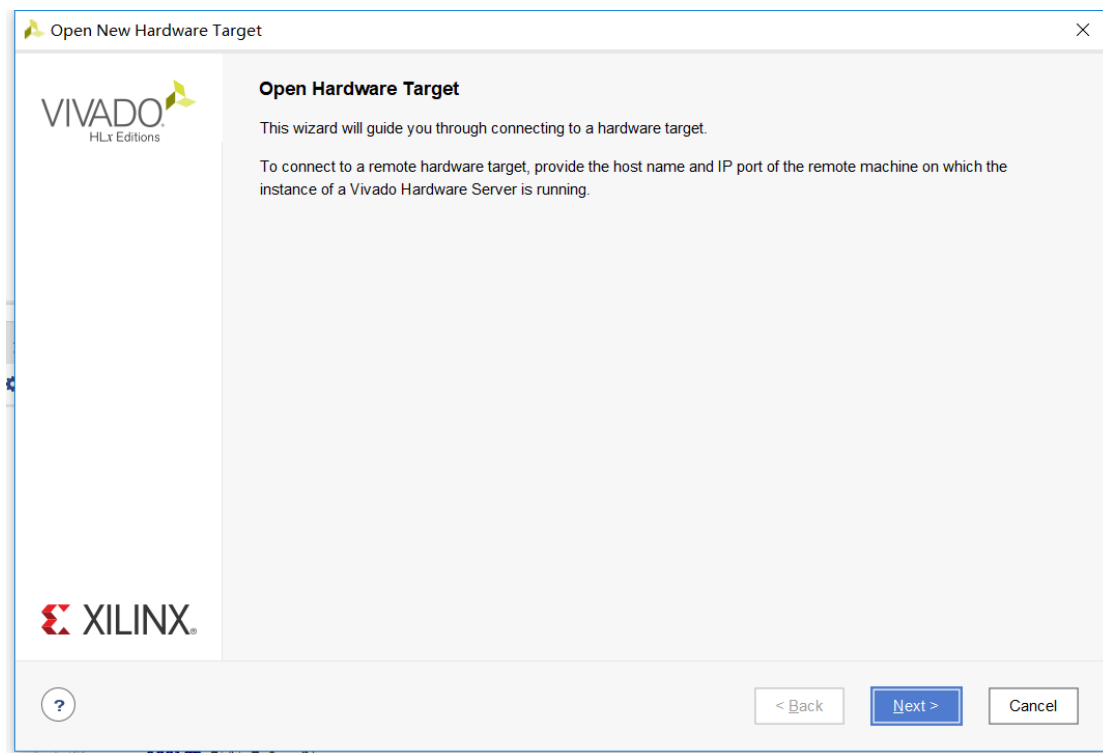
Link power and download cable through a Micro B USB connector, **J17**.and open power switch.



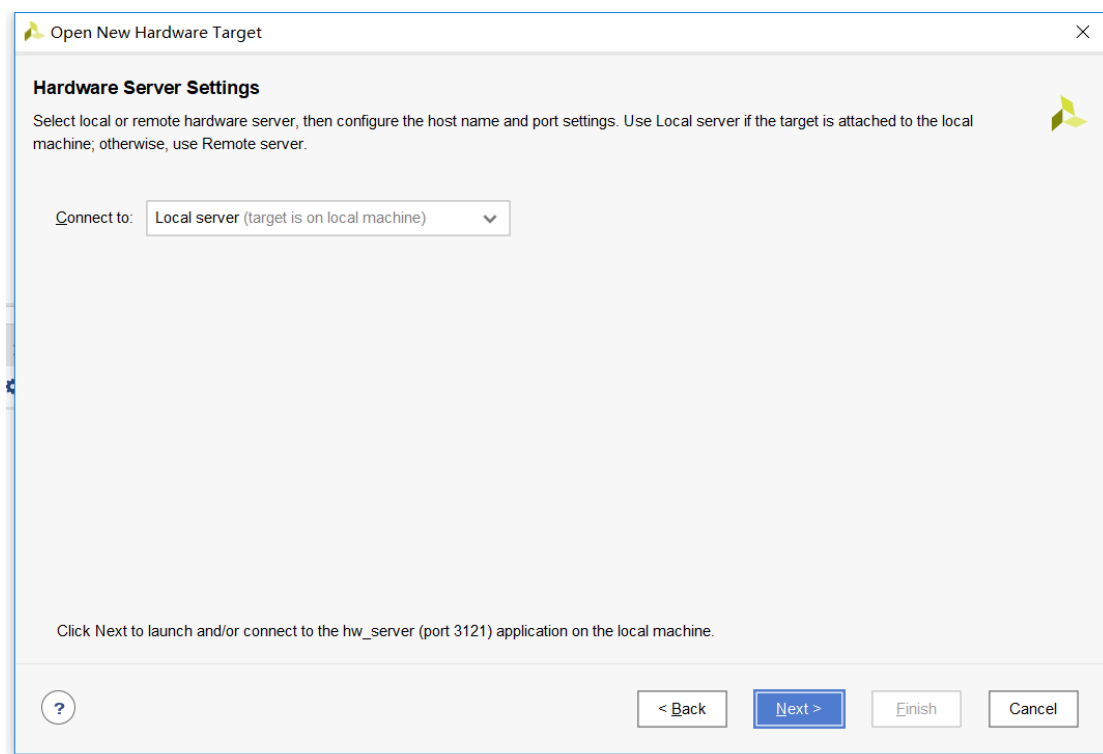
Click “Open target” and select “Open New Target”



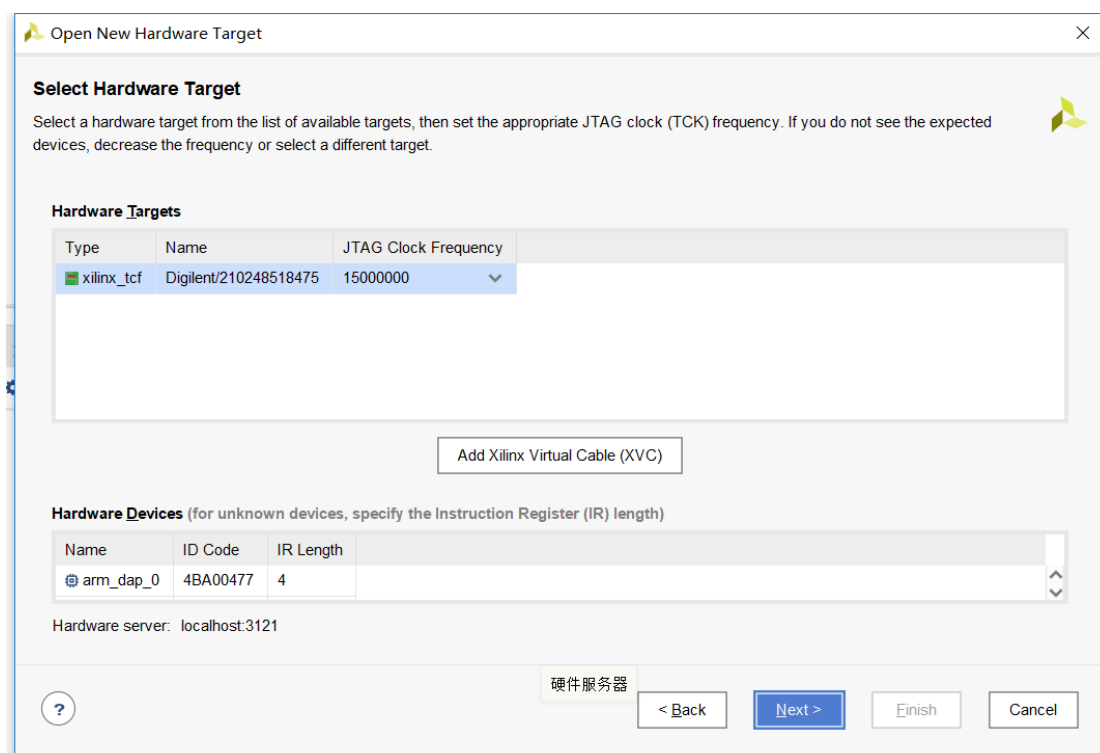
Click “Next”



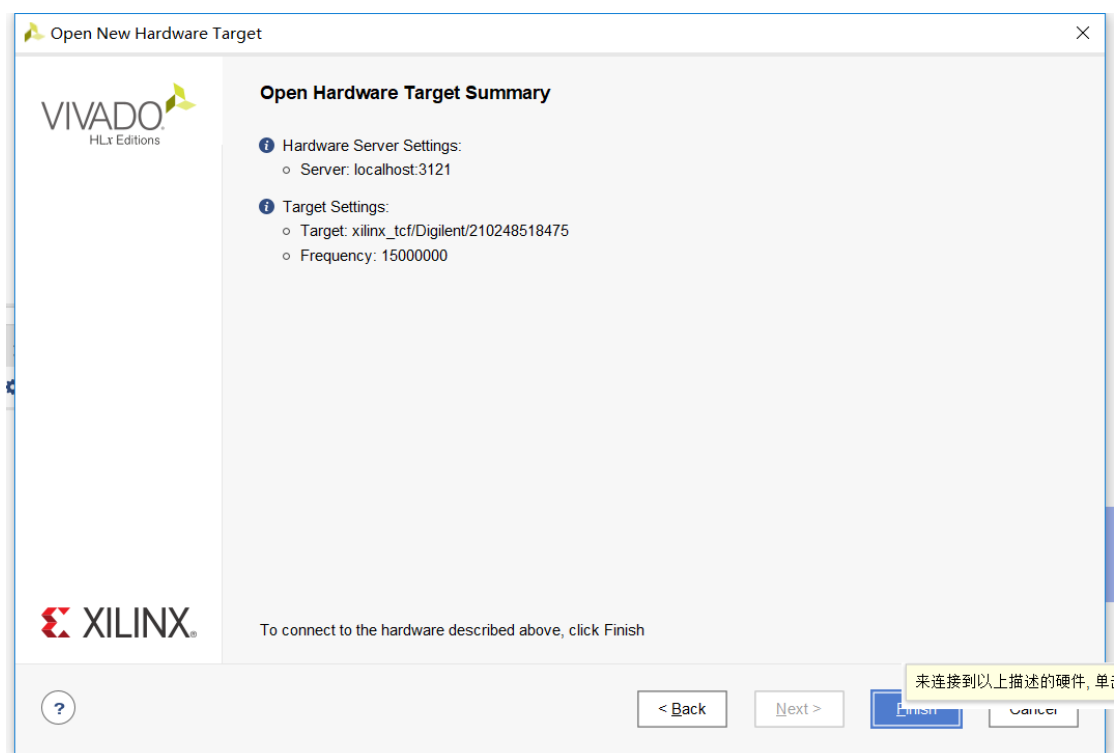
Click “Next”



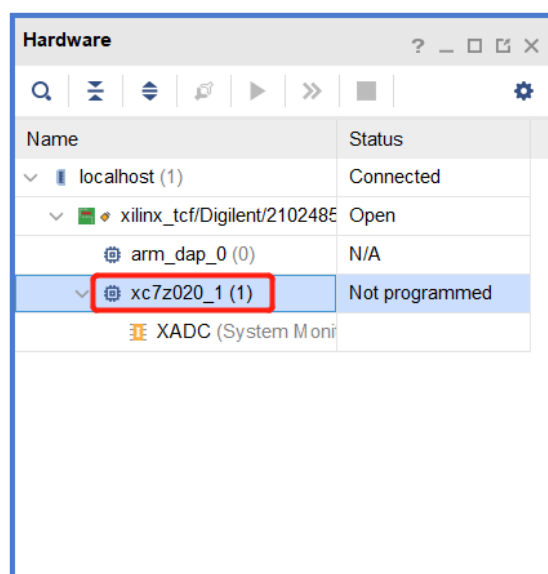
Click “Next”



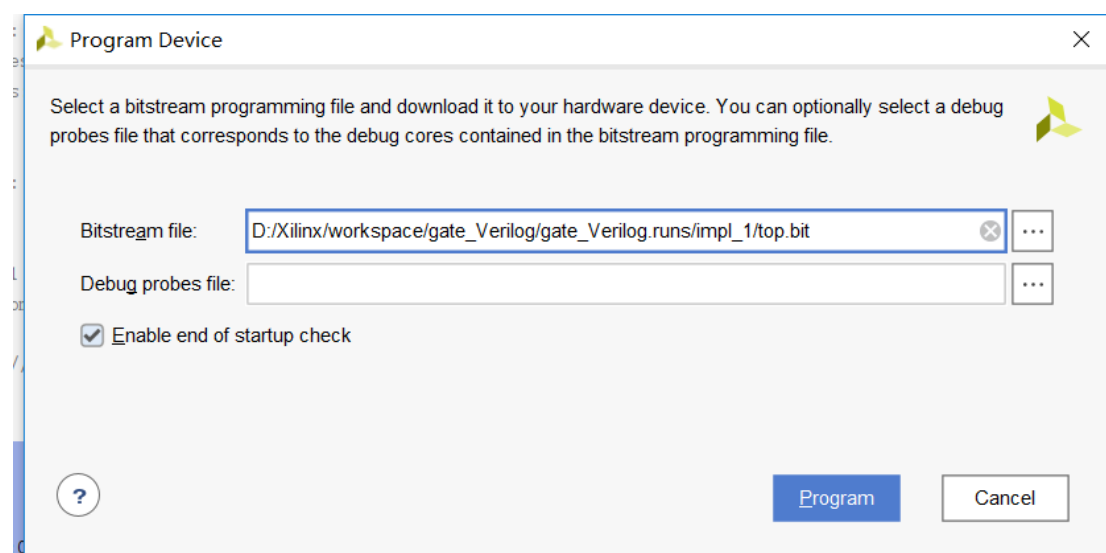
Click **“Finish”**



Right click device **“xc7z020\_1”**, select **“Program Device”**



Click **“Program”**

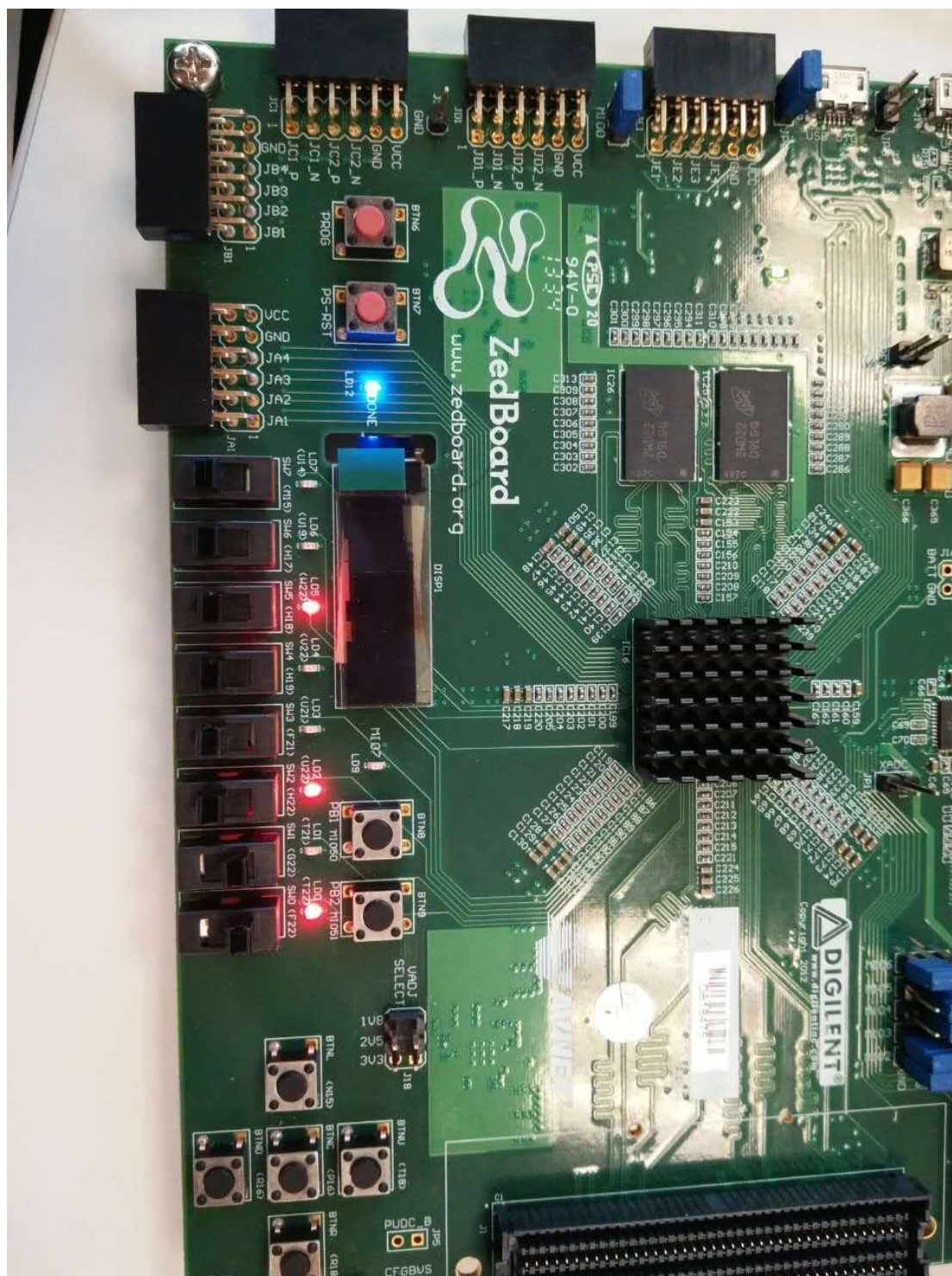


Verify the correctness by the switch sw1,sw0 through the board, compared to the truth table

Truth Table

| a | b | Z[5:0] |
|---|---|--------|
| 0 | 0 | 101010 |
| 0 | 1 | 010110 |
| 1 | 0 | 010110 |
| 1 | 1 | 100101 |





## Exercise:

According to previous memory、clock divider and address counter lab, now you need to integrate this lab and download to Zedboard using upper tutorial. Here we need clock source from pin Y9 on the basis of manual.

### 2.5 Clock sources

The Zynq-7000 AP SoC's PS subsystem uses a dedicated 33.3333 MHz clock source, IC18, Fox 767-33.33333-12, with series termination. The PS infrastructure can generate up to four PLL-based clocks for the PL system. An on-board 100 MHz oscillator, IC17, Fox 767-100-136, supplies the PL subsystem clock input on bank 13, pin Y9.

